# The ground truth fails its own physics check: the cost of auditing a CFD surrogate with an operator-inconsistent residual

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## Abstract

Machine-learning CFD surrogates predict steady flow fields far faster than classical solvers; the standard proposal for trusting them is a self-check against the governing equations. We show that this check rests on an operator the truth itself cannot pass. On 200/200 real AirfRANS cases the monitored residual of the *ground-truth* field is nonzero (mean 0.192), while the physically wrong uniform freestream scores an exact zero: the objective prefers a wrong field to the truth. This is not a resolution artifact—the monitored operator is an inconsistent discretisation, so the floor has a nonzero continuum limit, and two independent implementations measure it *growing* under refinement: rising in 21 of 24 cases on one, decaying in 0 of 16 on the other, by a pre-registered rule. The consequence for correction is direct: started at the exact ground truth, with no network in the loop, line-searched gradient descent on the residual drives the field error up on 200/200 cases, and on 94–98% from the deployed surrogate across three seeds ($p < 10^{-30}$), unchanged under three step rules. Descent does not always harm—from a perturbed start it often helps before overshooting—but the residual never selects the error-optimal iterate, failing to in 24/24 cases on both arms. What the monitor does deliver is triage and a certificate: it flags worst-decile drag error at AUROC 0.952 and supports a valid split-conformal band at 0.89 coverage, at a width 5.6–6.8× the drag error it bounds.

**Keywords:** neural operators, surrogate CFD, physics residuals, uncertainty quantification, conformal prediction, trust calibration, airfoil aerodynamics.

## 1 Introduction

Classical CFD resolves the governing PDEs by iterating a discretised system until its residuals fall below a tolerance. It is accurate and trusted, but slow: meshing and a converged steady-RANS solve over a new geometry can take minutes to hours, throttling the rapid design-space exploration that early engineering most needs. Neural surrogates promise the opposite trade—millisecond inference—and a vigorous literature now maps geometry plus boundary conditions directly to flow fields with neural operators and attention models.

The problem is **trust**. A trained surrogate is a one-shot function: it emits a field with no self-assessment, no convergence signal, and no recovery path when it extrapolates. On a geometry far from the training set—the common case in design—the user cannot distinguish a faithful prediction from a confident hallucination.

The governing physics offers an obvious lever. The steady incompressible RANS residual of any candidate field is computable directly from the field, with no ground truth. A neural-CFD pipeline could use this residual in three distinct ways: as a *trust signal* that flags *which* predictions to distrust (a case-level error proxy that drives a conformal trust layer), as a *correction objective* that a step should minimise, and as an *acceptance gate* that a step must not worsen. These are not the same role, and—this is the crux—they do not earn the same verdict. The measurement that decides it is stark: on 200/200 real AirfRANS cases the monitored residual of the *ground-truth* field is nonzero, while the physically wrong uniform freestream scores an exact zero. Put plainly, the residual tells you *which* predictions are wrong, not *how* to fix them. This paper separates the two roles and answers each empirically on a SOTA backbone, and—separately—deploys a learned, supervised corrector whose gain we attribute by ablation to the learned correction, not to the residual.

**Research question.** *A deployed surrogate cannot afford the solver that produced its training data, so any self-audit must monitor a cheaper, different operator. What does that substitution cost, and which of the residual's roles—error detector, correction objective, acceptance gate—survive it?*

**Thesis (one sentence).** ***The steady-RANS residual a deployed surrogate can afford to monitor is inconsistent with the operator that generated its labels, so the ground truth itself fails the check by a margin that grows under grid refinement; consequently the residual cannot be minimised—the iterate it selects is never the error-optimal one—while it remains useful for ranking errors, and the boundary between the residual corrections that work in the literature and the failure we report is operator consistency, not the problem class.***

**Why this is not a statement about our grid.** The obvious reply to a nonzero residual at the truth is that the grid is too coarse. We test it directly and it is wrong: refining from $128^2$ to $512^2$ makes the floor *larger*, in 21 of 24 cases, by a rule registered before the measurement (subsection 5.6); a second, independent 16-case, five-rung implementation finds the floor decaying in 0 of 16 cases, at $p = -0.64 \pm 0.29$, with its own reproduction gate passing to $3 \times 10^{-8}$ relative (section 6). The sharpest control is on the real data rather than a manufactured one: rasterising the *same* reference cloud with cubic instead of linear interpolation *raises* the floor by 6–12%, and lowers it in only 2–4 of 16 cases, so the kinks of a piecewise linear reconstruction are not what sets it. Restoring the omitted closure term moves it by less than 0.2%. The uniform freestream, meanwhile, scores an exact zero at every resolution, because a spatially constant field has identically zero finite differences—so the gap by which the objective prefers a wrong field to the truth widens as the grid improves.

**Self-auditing, defined—and what the physics earns.** We call a surrogate *self-auditing* if, from its own prediction and inputs alone—no ground truth, no reference solver—it supplies three measurable things: **(i)** a case-level *trust score* whose ranking of predictions by true error is validated; **(ii)** a *calibrated error band* with a distribution-free guarantee; and **(iii)** an *accept/reject decision* whose efficiency is measured against an oracle. We instantiate all three, and separate which of them the physics earns. (i) holds, but a *physics-free* ensemble score ranks field error at least as well and a paired bootstrap cannot separate them; the physics wins outright only on drag (subsection 5.10). (ii) holds outright, and needs no operator at all—though what the operator-based score can certify, it certifies only to a width the floor sets (subsection 5.9). (iii) holds as a *guarantee*—the gate cannot raise the residual—but its accuracy advantage does not survive an ungated fixed half step, so we claim the certificate and not the accuracy (subsection 5.5). What the residual uniquely supplies is none of these: it is the diagnosis of why the fourth role, correction objective, cannot work. In the shorthand we use throughout: the monitor can be *ranked* on, can *certify* only to a floor-limited width, and cannot be *descended*.

The evidence is a full predict→verify→estimate-uncertainty→correct→fall-back pipeline with pre-registered hypotheses: a 3-seed ablation on real AirfRANS, an out-of-distribution study, two formal certificates, and the same trust layer and corrector on a SOTA Transolver backbone. The verdict on the objective rests on minimising the residual directly, with no network in the loop (subsection 5.7), and takes the form the data supports: not that residual descent always harms the field—from a perturbed start it often helps—but that the iterate it selects is never the best one on its own path. Two findings accompany the learned corrector: correction quality is backbone-dependent, and a controlled ablation attributes its gain to the learned correction rather than to conditioning on the residual.

### 1.1 Contributions

In priority order:

1. **The ground truth fails the check (headline).** The residual a deployed surrogate can afford to monitor is not the operator that produced its training labels, and the gap is not small. On 200/200 real AirfRANS test cases the monitored residual of the *ground-truth* field is substantially nonzero—$\|r^\star\|$ mean 0.192, median 0.133—while the physically wrong uniform-freestream field scores an **exact zero** at every resolution, because a spatially constant field has identically zero finite differences. The objective therefore strictly prefers a wrong field to the truth (section 6). This is a property of the *operator*, not of any model: no network appears anywhere in the measurement.

2. **The floor cannot be refined away, and in practice it grows.** Two separate statements. *Provably*, the monitored operator is an inconsistent discretisation of RANS—it uses $\nu_{\text{eff}}\nabla^2 u$ in place of $\nabla\cdot(\nu_{\text{eff}}(\nabla u + \nabla u^\top))$—so its floor has a nonzero continuum limit rather than converging to zero as $h \to 0$, so (H2) stops being an assumption (section 6). *Empirically*, the floor does not merely persist but **rises** under refinement: over 24 cases at $128^2/256^2/512^2$, scored on a region fixed in physical units, $\|r^\star\|$ goes $0.0624 \to 0.0779 \to 0.1067$, rising in 21/24 cases, fitting $\|r^\star\| \sim h^p$ with $p = -0.387$ (subsection 5.6). The decision rule was registered before the measurement. The omitted closure term grows as predicted but accounts for only 3.5–4.8% of the floor: restoring it moves each level's mean by $-0.01\%$ to $+0.18\%$, and no individual case by more than 1.5%. Neither a higher-order interpolant (cubic *raises* the floor by 6–12%) nor operator repair removes it. What sets its size is the mismatch between our Cartesian stencil and the body-fitted discrete balance the reference solved.

3. **Descending the residual moves the field away from the truth, and the residual-selected iterate is never the best one.** We test this directly, by minimising the residual with the differentiable operator and no network in the loop. Under Armijo-line-searched gradient descent—so the objective is monotone non-increasing by construction—on 200 AirfRANS cases, starting at the *exact ground truth* drives the field error up on **200/200** cases, ending at 1.57× the entire error of the deployed surrogate; starting from that deployed surrogate it drives the error up on **94**–**98**% of cases across three seeds (median +74 to +87%, $p < 10^{-30}$), and the verdict is unchanged under three step rules and twelve decades of step size (subsection 5.7). We state the claim as one about **iterate selection**, because that is what holds universally: the error-minimising and residual-minimising iterates differ in 24/24 cases on both a truth-started and a perturbed-start arm, and the residual-selected iterate is strictly worse in 24/24. We explicitly do *not* claim residual descent always harms. From a perturbed start it *reduces* error in 18/24 cases before overshooting, and on a backbone far from the floor it lowers error on 75% of cases—so this is a regime claim, and the deployed backbone already sits inside the regime where descent almost never helps.

4. **The boundary is the operator, not the problem class.** Residual-driven correction demonstrably succeeds on steady CFD when the residual is the *solver's own*: Lei et al. (2026) drive a surrogate initial guess with Newton–Krylov correction and lower the median residual ratio by seven orders of magnitude *while* reducing field and aerodynamic error. That is not a counterexample to our result, it locates it. Correction works where the monitored operator is solver-consistent, and fails where an affordable surrogate-side monitor is inconsistent with the data generator. We draw the boundary there rather than at steady-versus-transient or resolved-versus-under-resolved (subsection 5.8).

5. **The residual still ranks errors—but it earns its place only in fusion.** As a *detector* the residual is consistently positive across three architecturally distinct backbones ($\rho$ from 0.40 to 0.85, or 0.40 to 0.61 among the arms trained and evaluated at matched density—see the MeshGraphNet scope note in subsection 5.2) and flags worst-decile cases at AUROC 0.871. A *physics-free* ensemble $\sigma$ does at least as well (0.894), and a paired case-level bootstrap cannot separate them (+0.023, CI $[-0.035, +0.083]$); it is the *fusion* that reaches 0.905 and recovers ≈91% of the oracle. The one place physics wins outright is the engineering quantity: on $|\Delta C_D|$ the residual reaches AUROC 0.952 against the physics-free score, with bootstrap CIs excluding zero (subsection 5.10). We also confirm

the detector is not a difficulty proxy: conditioning on each case's own floor leaves partial $\rho = 0.561$ against a raw 0.610.

6. **What is deployable: a certificate, not an accuracy mechanism.** A distribution-free split-conformal layer attains target coverage (0.902$\pm$0.008 at 0.90), including under distribution shift. The monotone-residual acceptance gate admits a step on 99.8% of deployed cases and lowers true error on 89.3% of them—but an *ungated* fixed half-step does better on accuracy alone (95.8%, median 6.2%), so the gate's accuracy comes from damping the correction, not from testing the residual. What the gate uniquely buys is the **certificate**: an ungated step raises the monitored residual on 1.0% of deployed cases (8.8% on the ensemble path), and the gate makes that impossible by construction. On the deployed path that is one case in a hundred: a guarantee, not a frequent intervention (subsection 5.5).

7. **A reproducible, CPU-first open-source package** (`neuroforge-cfd`) with a frozen I/O contract, interchangeable backbones, a zero-download synthetic potential-flow data generator, an AirfRANS loader, the ablation/certificate harness, and the figures—so every claim runs on a laptop and every certificate is reproducible from a committed script.

We are explicit throughout about what is **measured** versus **assumed**, make no out-of-distribution coverage *guarantee*, and make a competitiveness claim only for the Transolver backbone our trust layer runs on (subsection 5.2), not for the grid backbone.

**How to read this paper.** section 3 defines the audit and proves its two properties—contraction under the acceptance gate, and split-conformal coverage—and section 6 proves the floor survives the continuum limit and characterises what the audit cannot detect at all. subsection 5.6 refines the grid and finds the floor grows, with the controls that rule out the operator, the interpolant and the pipeline; subsection 5.7 minimises the residual directly from the exact truth; subsection 5.8 draws the boundary at operator consistency. What the audit does deliver is in subsection 5.2 (detector across backbones, learned correction), subsection 5.10 (physics against a physics-free score), subsection 5.9 (calibrated bands), subsection 5.4 (three backbone families, two datasets, a regime shift) and subsection 5.11 (what it costs). Every number maps to a committed script and result file through `docs/REPRODUCE.md`, with a manifest recording seeds, environment and SHA-256 hashes. Readers who want only the central result can read subsection 5.6 and subsection 5.7.

## 2 Related Work

**Neural operators (FNO / Geo-FNO).** Fourier Neural Operators learn a resolution-agnostic mapping between function spaces by parameterising a global convolution in the spectral domain (Li et al., 2021). Geo-FNO (Li et al., 2023) extends this to irregular geometries via a learned deformation to a latent uniform grid where the FFT is valid. These are strong *one-shot* operators with no self-assessment; NeuroForge implements both as backbones and adds the trust layer around them.

**Point-cloud and physics-attention operators (DoMINO, Transolver).** DoMINO (Ranade et al., 2025) is a decomposable multi-scale point-cloud operator for external aerodynamics on large industrial meshes. Transolver (Wu et al., 2024) replaces quadratic attention with attention over learnable *physics slices*, giving linear cost and strong accuracy on PDE benchmarks; Transolver++ (Luo et al., 2025) scales it to million-scale meshes. We use Transolver in two ways: as a matched-budget baseline for our grid backbone (subsection 5.12), and as the SOTA backbone our trust layer and corrector actually run on (subsection 5.2), where the residual is a calibrated detector and a supervised learned corrector reduces field MSE. "SOTA" here means Transolver's standing as a published state-of-the-art architecture at the time of our experiments (ICML 2024); an independent 2026 benchmark still places Transolver(++) among the strongest aerodynamic surrogates (Scherz et al., 2026), while newer architectures have since advanced the AirfRANS accuracy frontier (Rigotti et al., 2026). Our claims concern what the monitored residual can and cannot do *on* a competitive backbone, not a leaderboard position.

**Learned solver-correctors and the "fixer" premise.** Several lines of work use the discretised residual or a coarse-solution error as a *correction objective*: learned PDE solvers with convergence guarantees (Hsieh et al., 2019), deep-equilibrium operators for steady PDEs (Marwah et al., 2023), iterative refiners (Lippe et al., 2023), learned residual-error correctors (Jha, 2024), and—most recently—training-free correction that solves a linearised inverse problem on the PDE residual at each rollout step, reporting up to 100× error reduction on transient benchmarks (Huang & Perdikaris, 2025). PINNs (Raissi et al., 2019) embed the residual in the training loss. NeuroForge implements a contractive DEQ corrector and a feed-forward residual-conditioned corrector squarely in this family. **Our contribution to this line is to test the residual-as-correction premise head-on and find it does not hold.** As a correction *objective*—where a step must reduce the residual—residual minimisation does not track error minimisation on a real RANS benchmark: descending $J = \frac{1}{2}\|R_h\|^2$ from the exact ground truth cuts the residual to 39.8% of its starting value while driving the field error up on 200/200 cases, and on 94–98% of cases started from the deployed backbone. And when the residual is fed as an input *feature* to a corrector supervised toward ground truth, a controlled ablation attributes the gain to the learned correction rather than to the residual input (subsection 5.2). The residual's value is as a *trust signal*—it ranks which predictions to distrust—not as a correction mechanism telling you *how* to fix them; the learned correction does the fixing. The prior successes and our negative are consistent, and what separates them is the *operator*, not the problem class. Residual-driven correction works where the monitored residual is the solver's own, hence (near-)exact at the true solution: resolved transient PDEs stepped on their training grid (Huang & Perdikaris, 2025), variational problems with certified residuals (Jha, 2024), and—on steady CFD, so the boundary is demonstrably not steady versus transient—a Newton–Krylov correction driven by the solver's own residual, which lowers it by seven orders of magnitude *while* reducing field and aerodynamic error (Lei et al., 2026). It fails where the monitor is instead a surrogate-side operator that a deployed system can afford, inconsistent with the one that produced the labels. Carrying *some* floor is not sufficient for that failure—every discrete monitor carries one—so the operative quantity is how much of a prediction's score the floor already accounts for. On the deployed arm the median truth-to-prediction residual ratio is 0.864: 86% **of a typical prediction's score is present at zero error**, a fraction set by the operator rather than the model, which prices the certificate rather than invalidating it (subsection 5.9). The floor is a necessary condition for the failure mode we report, and the prior successes operate where it is absent. Our negative is scoped to that deployment-realistic regime.

**Residual-based, data-free error signals.** Closest to our *trust-signal* half is recent work that reads the PDE residual as a reference-free indicator of prediction error. Roy et al. (2025) use a physics-informed, residual-based error estimator to detect and correct accumulating error in neural-operator *time-marching*, triggering a classical solver when the estimator fires; Song et al. (2026) design an epistemic-UQ scheme whose uncertainty bands are aligned with localised residual structure in operator surrogates. Gopakumar et al. (2025) go furthest in the trust direction: they use the PDE residual itself as a conformal nonconformity score, obtaining data-free coverage guarantees—in *residual* space, with the transfer to solution space left open as a set-propagation problem. Our residual-floor theorem (section 6) supplies a concrete mechanism for that gap in the discrete steady-RANS setting: the monitored residual has a nonzero floor at the reference solution and a kernel of undetectable modes, so residual-space smallness cannot by itself certify solution-space accuracy. The question we put to the residual—can a quantity computable without the true solution certify that solution's accuracy?—is the defining question of a-posteriori error estimation, and the classical answer is instructive: residual magnitude is not itself an error estimate, and becomes one only when weighted by the solution of an adjoint problem measuring how strongly each local residual propagates into the functional of interest (Becker & Rannacher, 2001). The neural analogue keeps that structure: Hillebrecht & Unger (2025) bound the true error of a PDE-defined PINN by its residual scaled by a stability constant of the underlying operator. Both results are conditional on a property our monitor does not have—that the residual vanishes at the exact solution—and the residual-floor theorem is precisely a statement of how that condition fails for an under-resolved discrete steady-RANS operator. That is why we use the residual to *rank* cases rather than to *bound* them, and obtain the bound from conformal calibration instead. From the theory side, Mukherjee et al. (2026) prove that vanishing residuals guarantee solution convergence only under additional compactness assumptions; our setting is the practical complement, where the discrete monitored residual provably does *not* vanish at the dataset truth. We share the premise that the residual is a data-free error signal and do not claim to originate it. Our distinct contributions are (i) the head-to-head *dissociation*—the residual is a good trust signal but a poor correction objective, established on the same steady-state RANS benchmark; (ii) the *residual-floor theorem* establishing a nonzero continuum limit for the floor and characterising the undetectable kernel modes; and (iii) a split-conformal certificate calibrated on the *deployed, corrected* field. Our setting is steady-state external aerodynamics with per-case spatial trust, rather than time-marching error control.

**Uncertainty quantification and conformal prediction for PDE surrogates.** Deep ensembles (Lakshminarayanan et al., 2017) and MC-dropout (Gal & Ghahramani, 2016) are standard epistemic estimators; conformal prediction gives distribution-free coverage and has been applied to operator learning (Ma et al., 2024). The same combination is being pursued from both the PINN and the operator side: Yu et al. (2026) conformalise the output of a physics-informed network, and Garg & Chakraborty (2025) give distribution-free bands for neural operators. Both calibrate the *raw* surrogate; we calibrate the field the user actually receives—the corrected one—which matters here because the correction step changes the very error distribution the quantile is fitted to (subsection 5.9). Concurrently with this work, Jia et al. (2026) develop multi-granularity conformal calibration (case-level quantile regression plus residual-normalised point-level bands) for neural-operator automotive surrogates on DrivAerML; their nonconformity scales are purely data-driven, whereas our trust signal is the physics residual itself, and we additionally test—and reject—its use as a correction objective. We do **not** propose a new UQ method. Our trust layer applies split-conformal calibration to MC-dropout uncertainty; the new content is the *characterisation* of when this holds for CFD surrogates—in particular that coverage survives out-of-distribution empirically via uncertainty inflation, while the formal guarantee does not transfer once exchangeability fails.

**Benchmarks (AirfRANS).** AirfRANS (Bonnet et al., 2022) provides $\sim$ 1000 incompressible steady-RANS simulations over NACA airfoils with `full`, `scarce`, `reynolds`, and `aoa` splits designed to probe generalisation. We use `full` for in-distribution evaluation and the regime-disjoint `reynolds`/`aoa` splits for the out-of-distribution study.

### 2.1 Positioning

The closest prior art treats the physics residual primarily as a *correction objective* (PINNs, FNO-DEQ, learned correctors) or treats UQ in isolation from physics. NeuroForge's distinctive question is which of two roles the residual can play, evaluated head-to-head on the same model and data and then on a SOTA backbone: as a trust *signal* (detect error, drive conformal calibration) and as a correction *objective*, with the acceptance *gate* built on it judged separately. Our answer—signal yes (it tells you *which* predictions to distrust), objective no, gate yes as a guarantee and not as an accuracy mechanism, and certification only at a floor-limited width—and the resulting calibrated trust layer are the contribution. Separately, a learned, supervised corrector reduces field error on the strong backbone, with a controlled ablation attributing that gain to the learned correction rather than to the residual input. The no-harm/contraction machinery of section 3 carries its own guarantee, and the acceptance gate is materially useful: it admits a step on 99.8% of deployed cases and that step lowers field error on 89.3% of them (subsection 5.5). Table 1 places the neighbouring work on the boundary this paper draws. The organising question is not which features a system has, but whether the residual it drives is the *same discrete operator* that produced its reference data. Where it is, residual-driven correction works and is already demonstrated. Where an affordable surrogate-side proxy stands in for it, the truth itself fails the check, and this paper is about what follows from that.

The trust-layer components this paper retains—distribution-free conformal bands (Gopakumar et al., 2025; Jia et al., 2026), residual-based error signals (Roy et al., 2025; Song et al., 2026) and a-posteriori certification (Mukherjee et al., 2026)—are individually well developed in that recent literature, and we claim no priority over them. What is new here is the measurement that constrains all of them: the operator such a layer can afford to monitor does not vanish on the data it is calibrated against, and that gap is not a resolution artifact.

Table 1: The boundary, and where the closest 2024–2026 work sits on it. The discriminating question is whether the residual being monitored or minimised is the discrete operator whose root the reference solution is. Where it is, the floor at the truth is zero by construction and minimising the residual is minimising the error. Where it is not—the affordable, deployment-time case, where a cheap proxy operator stands in for the generator's—the floor is nonzero and the objective's minimiser is displaced from the truth, which we test by explicitly minimising $J = \frac{1}{2}\|R_h\|^2$ at $n = 200$ under three step rules, not by observing a supervised corrector. Prior residual-correction successes and this paper's negative result are not in conflict; they sit on opposite sides of this line.

| | **Solver-consistent operator** | **Surrogate-side proxy operator** |
|---|---|---|
| Floor at the truth | Zero by construction. | Nonzero, with a nonzero continuum limit (section 6); it *grows* under refinement, $p = -0.387$, rising in 21/24 cases (subsection 5.6). |
| Residual as a correction objective | Works: residual ↓ and error ↓ together (Lei et al., 2026; Huang & Perdikaris, 2025; Jha, 2024). | Fails: descent from the exact truth drives the field error up on 200/200 cases under Armijo-line-searched gradient descent, and on 94–98% of cases started from the deployed backbone ($p < 10^{-30}$, unchanged under three step rules); the residual-selected iterate is never the error-optimal one, in 24/24 cases (subsection 5.7). |
| Residual as an error detector | Not the question these works ask. | Positive across backbones, but matched by a physics-free score on field error; it wins outright only on drag (subsection 5.10). |
| Cost at deployment | A solver-grade operator per correction step. | $\sim$1.1 ms per case (subsection 5.11), which is why it is the one a deployed surrogate can actually afford. |

## 3 Method

### 3.1 Pipeline overview

The NeuroForge framework wraps a one-shot backbone in a verify–estimate–correct–fall-back loop (Figure 1). Throughout, the monitored residual is asked to do three different jobs and earns **three different verdicts** (section 5). It can be *ranked* on: as a trust signal it detects error between cases and drives the conformal layer, positively across every backbone we test. It *certifies*, but only to a floor-limited width, because most of a typical score is already present at zero error. And it cannot be *descended*: as a correction objective it fails, though the *acceptance gate* built on it still delivers its guarantee (subsection 5.5). Separately, a learned, supervised corrector improves a SOTA backbone (subsection 5.2). We describe each component and flag which role it instantiates.

### 3.2 Geometry encoding and governing equations

A `FlowCase` (geometry + boundary conditions + fluid + domain) is encoded into a fixed channel-first stack $x \in \mathbb{R}^{7\times H\times W}$ in the frozen `INPUT_CHANNELS` order $(\texttt{sdf}, \texttt{mask}, x, y, u_{\text{in}}, v_{\text{in}}, \log \text{Re})$: the signed distance to the body surface (negative inside the solid), a fluid/solid mask, normalised cell coordinates, the freestream velocity broadcast over the grid, and $\log_{10}\text{Re}$. The backbone outputs `OUTPUT_CHANNELS` $(u, v, p, \nu_t)$, where $p$ is the **kinematic** pressure $p/\rho$.

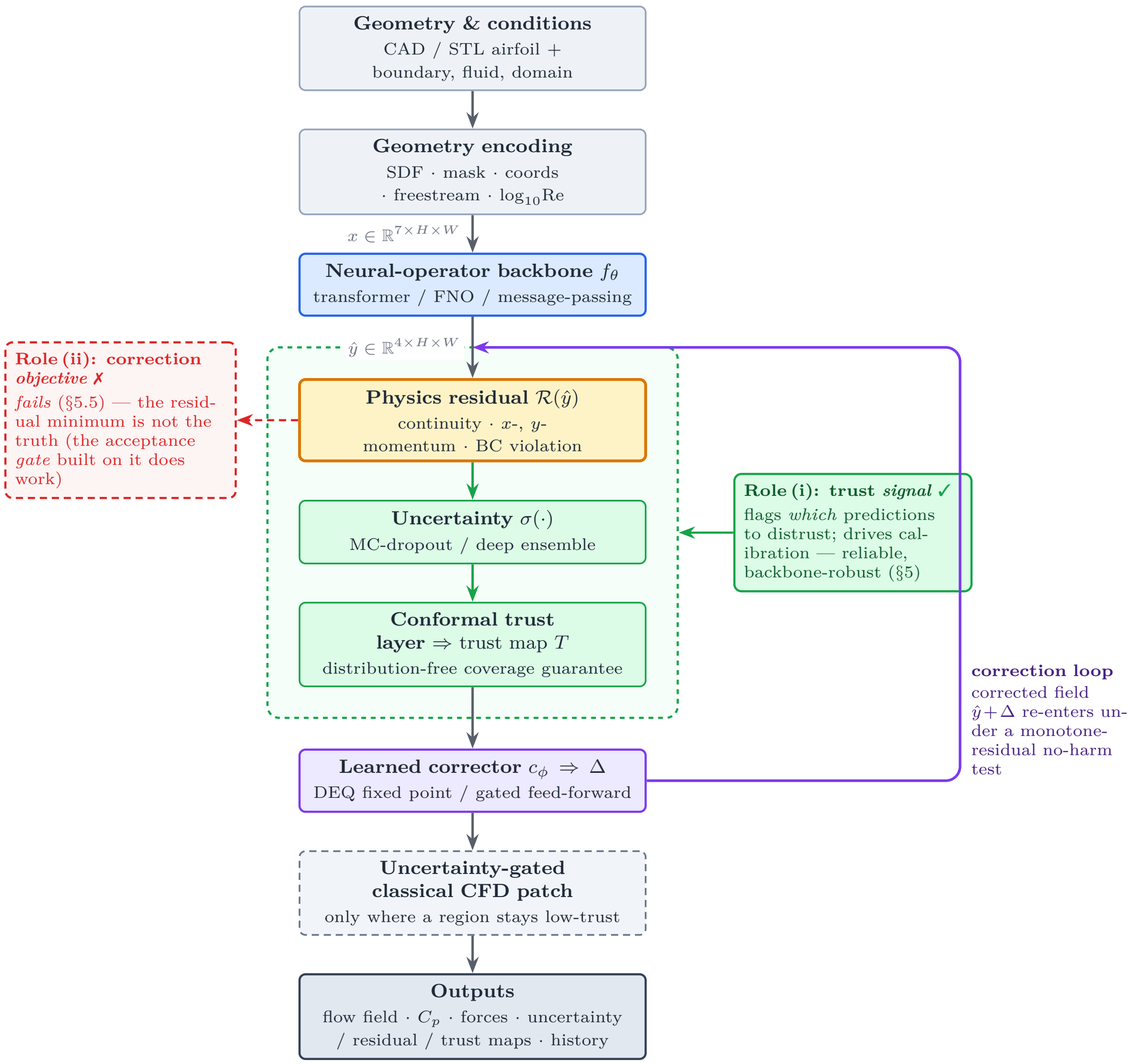


Figure 1: **The NeuroForge pipeline.** A one-shot backbone $f_\theta$ is wrapped in a verify–estimate–correct–fallback loop. Colour marks each component's role: the physics residual (amber) acts as a reliable *trust signal* (role i, green) but not as a *correction objective* (role ii, red); the deployed correction is a *learned* corrector (violet) applied under a monotone-residual no-harm test. The figure marks the two roles the residual plays as a *pipeline component*; the third verdict—that the conformal certificate it supports is valid but limited in width by the floor—is a property of the green block and is quantified in subsection 5.9. The roles, and the attribution of the corrector's gain, are established and scoped in the text (section 3, section 5).

The verifier evaluates the steady, incompressible, 2-D RANS equations in primitive form on the **physical (denormalised)** fields, with effective viscosity $\nu_{\text{eff}} = \nu + \nu_t$, pointwise:

$$\text{continuity:} \quad r_c = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}, \tag{1}$$

$$x\text{-momentum:} \quad r_x = u\,\frac{\partial u}{\partial x} + v\,\frac{\partial u}{\partial y} + \frac{\partial p}{\partial x} - \nu_{\text{eff}}\,\nabla^2 u, \tag{2}$$

$$y\text{-momentum:} \quad r_y = u\,\frac{\partial v}{\partial x} + v\,\frac{\partial v}{\partial y} + \frac{\partial p}{\partial y} - \nu_{\text{eff}}\,\nabla^2 v. \tag{3}$$

Because $p$ is kinematic, no explicit density appears. **The turbulence term is live.** Over the fluid cells of the 200-case AirfRANS test split the eddy viscosity averages $\nu_t = 8.2 \times 10^{-5}$ against a laminar $\nu = 1.56 \times 10^{-5}$: $\nu_t \approx 5.2\,\nu$, supplying $\approx 84\%$ of $\nu_{\text{eff}}$. The monitored operator is therefore a genuine RANS residual rather than a laminar one under another name, and the closure the surrogate predicts is what the audit consumes. The channel is learned, though backbone-dependently: against $\nu_t$'s own fluid-masked variance, $R^2 = 0.996$ on the grid backbone and 0.67–0.70 on MeshGraphNet (`scripts/check_nut_learned.py`). We report $R^2$ rather than raw MSE here deliberately: $\nu_t$ is three to four orders of magnitude smaller than $u$, so a per-channel MSE compared across channels understates its accuracy for purely dimensional reasons. What the operator *does* omit is the spatially varying term $\nabla\nu_t\cdot\nabla u$, which is one reason the residual floor of section 6 is nonzero. Derivatives use central finite differences with one-sided stencils at the borders; residuals are zeroed inside the solid. A boundary-condition violation map $r_{bc}$ adds a **no-slip** term penalising velocity magnitude in the thin fluid band adjacent to the wall (weighted by $\exp(-|\text{sdf}|/\ell)$, $\ell \approx 3$ cells) and a **far-field** term penalising deviation from $(u_\infty, v_\infty)$ on the outer border ring.

**Two implementation facts that bear on later claims.** *(i) The monitor discards its own strongest no-slip cells.* Residuals, including $r_{bc}$, are zeroed on the fluid ring immediately adjacent to the solid, because a central stencil there reaches into the $u = v = 0$ discontinuity and returns a spurious value an order of magnitude above the bulk. That masking is necessary, but it removes a mean 46% (range 35–51%) of the squared no-slip penalty—an unlogged spot check over 8 cases, which we report as such—precisely the band where the proximity weight is largest. Any claim resting on the *magnitude* of $r_{bc}$, including leg (A) of section 6, is therefore a claim about the surviving 54%, and we flag it there rather than let the weight profile suggest otherwise. *(ii) There are two Laplacian stencils in this pipeline.* The numpy monitor whose numbers we report uses the compact three-point form; the differentiable twin used to build the corrector's residual *input* channels uses a repeated first difference, which is wider and annihilates the period-2 mode the compact form sees. The training-time operator therefore has a strictly larger kernel than the monitored one. This is a live alternative explanation for the null result of the residual-input ablation (subsection 5.2) and we do not claim to have excluded it. Where it could have mattered most—the descent of subsection 5.7, which necessarily minimises the differentiable twin—we check both operators explicitly and both fall together in 24/24 cases, so that result is not an artifact of the wider stencil.

We stress the standard caveat that drives our whole study: a low residual is **necessary but not sufficient** for correctness—a smooth near-freestream field can have near-zero residual yet be entirely wrong—so the residual is a *consistency monitor*, and whether it tracks error is the empirical question section 5 answers.

### 3.3 Trust map and conformal calibration

**Raw trust map.** The combined PDE-residual magnitude $\rho_{\text{res}} = \sqrt{r_c^2 + r_x^2 + r_y^2 + r_{bc}^2}$ and a per-cell predictive uncertainty $\sigma$ are each mapped to $[0, 1]$ (using an absolute physical reference scale $s = U_\infty^2/L + U_\infty/L$ when available, else a robust 95th-percentile normalisation with an absolute floor) and fused,

$$e = \text{clip}\big(w_r\,\hat{r} + w_u\,\hat{\sigma},\ 0,\ 1\big), \qquad T = 1 - e, \tag{4}$$

with $w_r = 0.6$, $w_u = 0.4$, giving a traffic-light field (green $e < 0.15$, red $e > 0.45$, else yellow). Uncertainty $\sigma$ is the channel-mean standard deviation from a deep ensemble or MC-dropout.

**Case-level vs. per-cell (scope of the residual signal).** Our quantitative trust result is **case-level**: the field-mean residual ranks which whole predictions to distrust (per-case Spearman 0.611, subsection 5.2). The residual's *per-cell* spatial rank correlation with error *within* a field is weaker—0.22±0.06 (Spearman) though 0.60 (Pearson) on the dropout-FNO at $n$=15 (`results/control/percell_residual_error.json`), and 0.166±0.155 on the deployed ensemble-mean field at $n$=200. So the residual carries a coarse magnitude-level spatial signal (high-residual regions tend to be high-error) but is not a reliable per-cell error *rank*. The spatial signal is, however, **scale-dependent**: a systematic post-processing study on the deployed ensemble-mean fields ($n$=200; `scripts/percell_trust_localization.py`, `results/control/percell_localization.json`) shows that this raw cell-level rank correlation roughly *doubles* once the map is denoised to region scale—0.323±0.232 with 16×16-cell patch pooling (+0.157, improving 93% of cases; Gaussian smoothing at $\sigma$=4 cells gives +0.135)—and the pooled patches flag top-decile-error *regions* at AUROC 0.68–0.72. Dynamic-pressure normalisation does not help (+0.008). The scope is therefore: the residual is a *case-level* ranker and a *moderate region-level* locator at ∼8–16-cell granularity, not a per-cell error rank. We use it to flag *which* predictions to distrust, read the traffic-light map at patch scale, and leave per-cell claims to the conformal band below, which is the per-cell coverage *certificate*; it uses the calibrated $\sigma$, not the raw residual.

**Split-conformal calibration.** Raw $\sigma$ is uncalibrated, so a threshold on it carries no guarantee. We add split-conformal calibration (cf. Ma et al., 2024): on a held-out calibration set we compute per-channel nonconformity scores $s = |\hat{y} - y|/\sigma$ and take the finite-sample-corrected $(1 - \alpha)$ quantile $q$; the band $q \cdot \sigma$ then satisfies the distribution-free coverage guarantee in Proposition 1.

**Proposition 1** (Distribution-free coverage under exchangeability)**.** *If the calibration scores and a test score are exchangeable, then the split-conformal band $q \cdot \sigma$ satisfies $\mathbb{P}\big(|\hat{y} - y| \leq q\,\sigma\big) \geq 1 - \alpha$.*

This converts the trust threshold from a hand-picked constant into a band with a known in-distribution coverage level. subsection 5.9 reports the measured coverage, the coverage-vs-$\alpha$ sweep, and the out-of-distribution behaviour, with the exchangeability caveat made explicit.

### 3.4 The correction loop: supervised corrector, residual objective, and gate

The framework offers two correction operators and a backtracking acceptance test, and the distinction between them *is* the intellectual core of the paper. The DEQ corrector is trained by supervised regression toward ground truth and applied directly without any acceptance test; it ingests the residual as an input channel (its deployed architecture, below), but its gain is attributable to the learned correction rather than to that input (the W1 ablation, subsection 5.2). This is the path that helps on a SOTA backbone (subsection 5.2). The backtracking acceptance test uses the residual as a *gate*—a step is admitted only if it lowers the residual norm; measured, this path works, while residual *minimisation* as an objective is the one that fails (subsection 5.5). We describe both precisely so the contrast is interpretable.

**Feed-forward corrector + backtracking acceptance test.** A small residual CNN $c_\phi$ predicts an additive correction conditioned on the current field, its current 3-channel residual map, and the geometry: $\Delta_k = c_\phi(y_k, R(y_k), x)$. A candidate $y_{k+1} = y_k + s\,\Delta_k$ is accepted only if it does not increase the residual norm $N(y) = \sqrt{r_c^2 + r_x^2 + r_y^2}$:

$$N(y_k + s\,\Delta_k) \leq N(y_k) + \varepsilon, \tag{5}$$

with $s$ halved up to four times on failure. By construction $N(y_0) \geq N(y_1) \geq \cdots \geq N(y_K)$—the residual norm is monotone non-increasing across accepted steps. This is the convergence discipline of a classical solver. Its weakness is exactly the one we exploit as a finding: the test is satisfiable by the identity (zero step), so a corrector that cannot reduce the residual is simply rejected. Empirically (subsection 5.5) the test rarely refuses outright: backtracking finds an admissible damped step on 99.8% of deployed cases, and that step lowers true error. It throttles rather than blocks—even though reducing the field error and reducing the PDE residual are not the same objective.

**Contractive DEQ corrector.** To give the loop a genuine convergence guarantee in its *own* iteration variable, the correction $\delta$ is defined as the fixed point of a learned operator

$$\delta^* = T_\theta(\delta^*; c), \quad T_\theta(\delta; c) = \kappa \cdot g_\theta([\delta, c]), \quad c = [\hat{y}, r(\hat{y}), \mathrm{geom}], \tag{6}$$

where $g_\theta$ is a CNN whose layers are spectrally normalised (each $\leq$ 1-Lipschitz) and $\kappa < 1$.

**Proposition 2** (Banach contraction of the corrector)**.** *With $g_\theta$ each-layer 1-Lipschitz and $\kappa < 1$, $T_\theta$ is a $\kappa$-contraction in $\delta$. By the Banach fixed-point theorem the equilibrium $\delta^*$ exists, is unique, and the iteration converges geometrically, $\|\delta_k - \delta^*\| \leq \kappa^k \|\delta_0 - \delta^*\|$.*

The construction underlying Proposition 2 is a Deep-Equilibrium model (Bai et al., 2019) with the Lipschitz constant controlled by spectral normalisation (Winston & Kolter, 2020), trained with Jacobian-Free Backpropagation (Fung et al., 2022). Crucially, $g_\theta$ is trained on **data** (it targets the correction toward truth, with the residual as an input *feature*) rather than by minimising the residual; at inference the converged $\delta^*$ is applied directly, **without** the backtracking acceptance test. We verify this contraction empirically (subsection 5.9: measured factor $0.78 < \kappa = 0.9$). Two scopes must be kept distinct, and subsection 5.5 leans on the distinction: the DEQ contraction is in the *correction variable* $\delta$, whereas the *PDE residual of the output field* is a different quantity that, as we show, can rise even as $\delta$ converges and the field error falls.

The feed-forward corrector is trained (backbone frozen) so that $c_\phi(\hat{y}_{\mathrm{norm}}, R_{\mathrm{norm}}, x) \approx y^\star_{\mathrm{norm}} - \hat{y}_{\mathrm{norm}}$, with the final convolution initialised near zero for a stable first iteration.

### 3.5 Uncertainty-gated classical fallback

After the loop, if the maximum uncertainty exceeds a threshold, the low-trust fluid region is handed to a `ClassicalFallback`. In the present release this is an **interface with a `stub` backend** that reports what would run (and the region size) without invoking an external solver; the `openfoam`/`su2` backends raise `NotImplementedError` with setup guidance. This keeps the package importable and runnable with nothing installed while fixing the integration seam; it is not part of any quantitative claim.

## 4 Implementation

NeuroForge is an open-source, **CPU-first**, pure-Python package (`neuroforge-cfd`, Python $\geq$ 3.10, NumPy/SciPy/PyTorch/Matplotlib). Importing the package caps BLAS/OMP/MKL thread counts to one by default to avoid catastrophic oversubscription on low-core machines, and does no heavy work. The codebase is organised around a **frozen I/O contract** in `core/`.

**Module map.** `core/` holds the frozen data contracts (`FlowCase`, `FlowField`, `Diagnostics`, `SolveResult`) and `Config`. `geometry/` builds the 7-channel input (NACA generation, SDF/mask rasterisation, `encode_case`). `data/` has the synthetic generator, the AirfRANS loader, the rasteriser, and the `datamodule` (`Normalizer`/loaders). `models/` provides `FNO2d`, `GeoFNO`, a Transolver-style `PhysicsTransformer`, `UNet`/`DeepONet` baselines, `LocalCorrectionNet`, the `DEQCorrector`, and the `DeepEnsemble`/`MCDropoutUQ` wrappers, all in a string-keyed registry. `physics/` has the differential operators, the `PhysicsChecker`, `trust_map`, force/Cp/error metrics, the conformal calibration, and the differentiable `physics_residual_torch`. `solver/` has `Predictor`, `NeuroForgeEngine`, `neural_residual_iteration`, and `ClassicalFallback`. `train/`, `viz/`, `cli.py`, `app/` provide training, plotting/report, the CLI, and a Streamlit UI.

**Fixed I/O contract.** Inputs are always the 7 channels $(\mathtt{sdf}, \mathtt{mask}, x, y, u_{\mathrm{in}}, v_{\mathrm{in}}, \log \mathrm{Re})$; outputs always the 4 channels $(u, v, p, \nu_t)$. Fields are `(ny, nx)` `float32`; network tensors channel-first $(B, C, \mathtt{ny}, \mathtt{nx})$. Pressure is kinematic; residuals are computed on denormalised fields with $\nu_{\mathrm{eff}} = \nu + \nu_t$. Fixing this contract is what makes the backbones interchangeable and the trust layer backbone-robust.

**Backbones.** `FNO2d` is a faithful spectral FNO; `GeoFNO` adds a geometry-gated conditioning of the lifted features; `PhysicsTransformer` implements Transolver-style physics-slice attention (linear in grid points); `UNet`/`DeepONet` are baselines.

**Training loss.** The `CompositeLoss` sums a masked data MSE over fluid cells, a differentiable physics-residual term on the denormalised fields, and a no-slip BC term. The corrector is trained separately with the backbone frozen (subsection 3.4).

**Synthetic potential-flow generator (zero-download reproducibility).** `SyntheticRANS` superposes a Hess–Smith source-panel potential core (with a Kutta-enforcing bound vortex), an algebraic near-wall no-slip ramp and Gaussian wake deficit, kinematic Bernoulli pressure, and a mixing-length $\nu_t$, with desingularised kernels and light smoothing. The fields are physically plausible and continuity-respecting but **analytic**—a smoke-test/reproducibility substrate, not solver ground truth. **No quantitative claim in this paper rests on synthetic data**; all reported numbers (section 5) are on real AirfRANS.

**AirfRANS loader.** `load_airfrans` reads each simulation's point cloud, reconstructs the airfoil loop, rasterises the targets onto a structured crop, and returns (`FlowCase`, `FlowField`) pairs for all four splits.

**Use of AI-assisted development tools.** Substantial portions of the research software (model implementations, training/evaluation harnesses, and analysis scripts) were developed with the assistance of a generative-AI coding tool (Claude, Anthropic; Claude Opus 4.8 and Claude Fable 5 models, used through the Claude Code environment), operating under the authors' direction. The authors specified, reviewed, and validated all code; every reported number is produced by a committed script in the public repository and verified against an artifact manifest of SHA-256 hashes (`results/MANIFEST.json`), and the full test suite accompanies the release. The scientific questions, experimental design, pre-registered hypotheses, and interpretation of results are the authors' own.

## 5 Experiments

All quantitative results are on **real AirfRANS**. Unless noted, the in-distribution and out-of-distribution ablations use the pre-registered protocol (`benchmarks/ablation.py`, `docs/EXPERIMENTS.md`) over **3 seeds (0, 1, 2)**, reported as mean ± std (population std, `ddof=0`; sample std at $n = 3$ widens bars by $\approx 1.22\times$). Metrics follow the AirfRANS community protocol (`evaluate_cases`): per-channel volume MSE (lower is better, well-conditioned), surface-pressure MSE on the body, Spearman rank correlation of the force coefficients $\rho_{C_l}, \rho_{C_d}$ (closer to 1 is better—what early-design ranking needs; throughout the tables these correlate forces integrated from the predicted vs. the ground-truth field through our grid surface-integrator—a *field-to-field self-consistency* valid for the relative arm/condition comparisons, not agreement with AirfRANS's official force labels; see subsection 5.2), and `residual_error_spearman` ($> 0$ means a low residual tracks low error). The $\nu_t$ channel is omitted from the tables because its raw MSE is not comparable to the other channels ($\nu_t$ is three to four orders of magnitude smaller than $u$); measured against its own fluid-masked variance it is learned, backbone-dependently ($R^2 = 0.996$ grid backbone, 0.67–0.70 MeshGraphNet; `scripts/check_nut_learned.py`, and see section 7).

With $n = 3$ we report **per-seed sign-consistency** plus effect-size magnitude rather than p-values, which are meaningless at $n = 3$. An effect called "robust" below is sign-consistent across all 3 seeds with a large effect size and (where stated) non-overlapping per-seed distributions.

### 5.1 Setup

The backbone is an FNO trained on AirfRANS `full` (800 train / 200 test, 80 epochs). Four arms: `backbone`, `backbone (no physics loss)`, `backbone + local corrector` (feed-forward, with the acceptance test), and `backbone + DEQ corrector`. The certificate runs (subsection 5.9) use a dropout-enabled FNO (width 48, 4 layers, 20 modes, dropout 0.05) with a DEQ corrector ($\kappa = 0.9$, damping 0.5), trained $40 + 15$ epochs at resolution 128. The v2 phase (subsection 5.2) re-runs the trust layer and DEQ corrector on a SOTA Transolver backbone under the same protocol; subsection 5.3–5.9 are the weak-grid-backbone runs that establish the backbone-dependence finding.

### 5.2 Backbone-agnostic trust + learned correction on a SOTA backbone (v2)

We re-run the trust layer and the DEQ corrector on a **SOTA Transolver backbone** (7.35M parameters, 80 epochs, AirfRANS `full`, 5 seeds; corrector trained 20 epochs with the backbone frozen), the strongest model in our study. The protocol and scoring are identical to subsection 5.3; the sources are `results/v2/v2_results.json` (original 3-seed study) and `results/control/w1_capture.json` (identical recipe, extended to five seeds; the extension reproduces the 3-seed corrector deltas and $\rho_{C_l}$ within pre-registered tolerances). Table 2 reports the backbone alone and with the DEQ correction loop over all five seeds.

**Learned correction works on a strong backbone.** The DEQ corrector reduces volume field MSE on **all five seeds**: `mse_u` −8% (0.133 → 0.122), `mse_v` −21% (0.096 → 0.076), `mse_p` −25% (644 → 485). The per-

Table 2: SOTA Transolver backbone (7.35M params, 80 ep, AirfRANS `full`), alone vs. with the DEQ correction loop. 5 seeds, mean ± std (population std, `ddof=0`). Lower MSE is better; $\rho$ closer to 1 is better; resid↔err $\rho > 0$ supports the trust signal. Source: `results/control/w1_capture.json` (5-seed extension of `results/v2/v2_results.json`). See subsection 5.2.

| metric | `mse_u` | `mse_v` | `mse_p` | $C_l$ rel. err | $C_d$ rel. err | resid↔err $\rho$ |
|---|---|---|---|---|---|---|
| backbone | 0.133±0.017 | 0.096±0.006 | 644±63 | 5.7% | 7.6% | 0.611±0.054 |
| + DEQ loop | 0.122±0.015 | 0.076±0.006 | 485±39 | 5.5% | 6.8% | 0.612±0.046 |
| $\Delta$ | −8% | −21% | −25% | −0.2pp | −0.8pp | +0.002 |

seed reductions are sign-consistent down on all five seeds for every volume channel. **Surface-pressure** MSE is the one channel the corrector does *not* improve on this strong backbone: essentially flat at $9843 \to 9899$ (+0.6%; per-seed $-1/-1/+2/+2/+1\%$). Internal force-ranking consistency is held at $\rho_{C_l} = 0.999$, $\rho_{C_d}$ flat at 0.996, with relative force errors $C_l$ 5.7% → 5.5%, $C_d$ 7.6% → 6.8%. This is the corrector trained by supervised regression toward ground truth and applied without the acceptance test (subsection 3.4).

**What these force correlations measure—and what they do not.** Our $\rho_{C_l}/\rho_{C_d}$ (here and in every table) correlate the force integrated from the *predicted* field against the force integrated from the *ground-truth* field through the same grid surface-integrator: a field-to-field consistency, valid for the relative arm/condition comparisons we draw from it, but *not* agreement with AirfRANS's official OpenFOAM force labels (which our pipeline never loads). Recomputed against those official labels (`scripts/recompute_force_vs_official.py`; 200 cases, 3 seeds; `results/control/force_vs_official.json`), the deployed backbone recovers force *ranking*—$\rho_D = 0.84{\pm}0.01$ (drag), $\rho_L = 0.88$ (lift)—but not magnitude: absolute drag is biased ($\approx 11\times$ mean relative error) by the near-field integrator's 1.5-cell surface-sampling offset, and the same integrator on *perfect* fields also yields $\rho_D = 0.84$, so this number is not limited by the prediction. Nor is it limited only by the measurement: a field-free covariate baseline beats it (below).

**That 0.84 is not a ceiling: a field-free covariate baseline beats it.** A three-parameter ordinary least-squares regression on $(U, \alpha, \alpha^2)$ that never touches a flow field ranks the official drag at $\rho = 0.874$—above every integrator arm we tried on *ground-truth* fields (0.611–0.839) and above every arm on predicted fields (0.828–0.845); $\alpha$ alone gives 0.800 (`scripts/drag_covariate_control.py`, `results/control/drag_covariate_control.json`). The operating conditions are recoverable because `Simulation.reset()` parses them from the case name. The consequence generalises beyond this paper: **a drag rank correlation reported on this benchmark is uninterpretable without its covariate baseline**, and the conventional near-field number sits below that baseline.

**What the rasterisation costs, measured against AirfRANS's own integrator.** To separate information destroyed by the raster from our choice of quadrature, we rasterise the exact truth to $128^2$, scatter it back to the source mesh nodes, and integrate with AirfRANS's *own* `force_coefficient` (`scripts/drag_observability_roundtrip.py`; four pre-registered gates all exactly 0.0). That round trip gives $\rho_D = 0.700$ and $\rho_L = 0.9967$—drag *below* our own 0.839, so the raster, not the quadrature, owns the loss. Controlling for the covariates leaves partial $\rho_D = 0.133$ against partial $\rho_L = 0.976$. The mechanism is resolution: 68% of drag is viscous and the wall shear lives in a sublayer at $y^+ \approx 5$, so recovering it needs $N \sim 10^5$ where lift, set by chord-scale surface pressure, converges at first order over the same ladder. Lift is observable from a raster at deployed resolution; drag is not, and the requirement is quantified rather than absolute.

**Repairing the measurement.** We then repaired what is repairable (`scripts/design_force_integrator.py`, `scripts/recompute_force_multischeme.py`; `results/control/force_vs_official_multischeme.json`). A wall-extrapolated pressure scheme cuts the ground-truth-field lift magnitude error from 16% to 4.6% (median), and a *control-volume* (far-field momentum-balance) integrator over the outer grid ring recovers official lift from ground-truth fields almost exactly ($\rho_L = 1.00$, median error $< 1\%$) and—the deployment-relevant number—from *predicted* fields at $\rho_L = 0.998 \pm 0.001$ with per-seed median magnitude errors of 3.6–3.9%: engineering-grade lift from the surrogate. Drag decomposes diagnostically: on ground-truth fields the control-volume estimator reduces the median drag error from 3.3× to 0.23× (the *measurement* is fixable), but on predicted fields it reaches only ≈ 2.7× with a degraded ranking, so the remaining drag error is attributable to the *model's* far-field/wake accuracy, not the integrator; drag *ranking* is still best served *among our integrators* by the near-field scheme ($\rho_D = 0.84$), which is nonetheless below the 0.874 covariate baseline and should not be read as a drag-ranking result. We therefore report lift as quantitatively recovered, read $\rho_{C_l}/\rho_{C_d}$ in the tables as internal consistency, and still make no claim of accurate absolute *drag* or of a cross-method force-ranking win (published methods use a different force-integration convention).

**Two limits on these force numbers.** $\rho_L = 0.998$ is an easy number twice over, and should not be read as evidence that the surrogate has learned the near-wall flow. Angle of attack is an *input* channel, and lift at these conditions is close to a function of it; separately, the control-volume estimator recovers lift from circulation through the outer ring via Kutta–Joukowski, so it depends on the far field the surrogate predicts most easily and not on the surface at all. The high correlation is real and useful for triage, but it is not a strong test. Second, and more restrictive: with residuals and forces evaluated on a grid whose solid-adjacent ring is masked, at $Re \approx 10^6$ the momentum thickness is far below one cell, so the *viscous* component of drag is not merely inaccurate here—it is **unavailable**. Every drag statement in this paper is therefore about the pressure-dominated component on a grid that cannot resolve the boundary layer, which is also why we do not report $C_{d,v}$ anywhere.

**The gain comes from the learned correction, not from feeding it the residual (W1 control).** The corrector improves the field by $-8/-21/-25\%$ (gate-verified). To isolate *why*, we train the same DEQ corrector two ways under identical architecture, data, optimiser, schedule and seeds, differing only in whether the 3 residual input channels carry the real residual (WITH) or are permanently zeroed at both train and inference (NULL). Across 5 seeds, the null corrector **matches or slightly exceeds** the deployed one: the per-seed paired difference $d = \text{gain}_{\text{WITH}} - \text{gain}_{\text{NULL}}$ is negative on all 5 seeds for both `mse_u` and `mse_speed` (mean $d \approx -0.005$ on a $\approx 0.12$ base, ≈ 4%; WITH beats NULL on 0/5 seeds; `results/control/w1_capture.json`). The residual input is therefore *not* the source of the corrector's field-MSE improvement; the improvement is attributable to the learned correction architecture, not to residual-conditioning. This is consistent with the residual's role elsewhere in the paper: it tells you *which* predictions are wrong (a trust signal), not *how* to fix them. The deployed system keeps the residual input because the residual is already computed for the trust layer and the acceptance test, so conditioning the corrector on it costs nothing extra; a residual-free corrector is an equally valid deployment. One caveat qualifies the null (section 3): those input channels are built from the wider differentiable stencil, whose larger kernel is a live alternative explanation we have not excluded.

**The trust signal is backbone-robust.** On this SOTA backbone the bare residual–error correlation is already strong (0.611±0.054 over five seeds—per-seed 0.611/0.652/0.613/0.667/0.511, positive on every seed; the wider $n$=5 spread is the error bar), vs ≈ 0.40 for the bare weak backbone in Table 4; the corrector barely moves it (0.612±0.046), because the strong backbone's residual is already a good detector without correction. The lift-by-corrector effect we report in subsection 5.3 is a property of the *weak* backbone; on the strong backbone the signal is strong from the start. We add a third, architecturally-distinct data point: a message-passing GNN (MeshGraphNet, param-matched at 7.35M, 3 seeds, bare backbone), where the bare case-level residual–error Spearman correlation is 0.851±0.058 (`results/mgn/mgn_results.json`). Across three fundamentally different architectures—a spectral grid Geo-FNO (≈ 0.40 bare), an attention/point-cloud Transolver (0.611±0.054 bare), and a message-passing GNN MeshGraphNet (0.851±0.058 bare)—the correlation is consistently *positive*, though variable in strength (a 0.40–0.85 range). This is precisely what "backbone-robust" means here: the detector *works* on every architecture, not that its strength is uniform across them. (The high MeshGraphNet value partly reflects that model's wide error spread—large, dispersed errors are easier to rank by residual.) MeshGraphNet is run only as a bare backbone (no corrector and no conformal layer), so it speaks only to the trust-signal claim.

Table 3: Conformal calibration of the pressure channel on the SOTA Transolver backbone ($\alpha$ = 0.1, 100/100 cal/test split): MC-dropout vs. a deep ensemble ($M$=5). Both attain target coverage; only the deep ensemble yields an input-adaptive band (finite $q$, low ECE). MC-dropout: 3 seeds (mean±std); deep ensemble: single ensemble training (split-to-split spread over 20 calibration/test redraws: coverage 0.899±0.009, $q$ = 2.26±0.05; subsection 5.9). Sources: `results/v2/v2_results.json`, `results/uq_ensemble/uq_results.json`.

| $\sigma$ estimator | coverage (target 0.90) | $q$ | ECE (↓) | band |
|---|---|---|---|---|
| MC-dropout ($p$=0.05) | 0.902±0.008 | $\approx 1.6 \times 10^9$ | $\approx 0.31$ | near-constant |
| deep ensemble ($M$=5) | 0.915 | 2.35 | **0.074** | **adaptive** |

**The MeshGraphNet arm is not a like-for-like third point.** That model was trained on 16k-point subsamples and evaluated at the full $\approx$ 180k points, and our own density control (`results/control/mgn_density_control.json`) returns the verdict DENSITY-DRIVEN, with an MSE ratio of 2.44 between the two densities. The 0.851 correlation is therefore measured on a model evaluated off its training distribution. The control measures MSE on four cases and carries no rank correlation, so it does not establish that $\rho = 0.851$ is *itself* a density artifact—but it is enough that the number should not be read as a like-for-like third point, and the 0.40–0.85 range should be read as 0.40–0.61 among the arms trained and evaluated at matched density. The detector remains positive on all three.

**Conformal coverage holds under both uncertainty estimators.** Split-conformal calibration of the pressure channel at $\alpha = 0.1$ attains target coverage on this SOTA backbone with *either* uncertainty estimator: 0.902±0.008 with MC-dropout (3 seeds; `results/v2/v2_results.json`) and 0.915 with a deep ensemble (single run; `results/uq_ensemble/uq_results.json`), both at the 0.90 target. Coverage was never the difficulty—**adaptivity** was. With MC-dropout this near-deterministic model has near-degenerate $\sigma$ at dropout 0.05, so coverage is delivered by a **near-constant** band: the conformal quantile $q$ is enormous ($\approx 1.5$–$1.7 \times 10^9$) and the reliability ECE is $\approx 0.31$.

**A deep ensemble recovers an adaptive band on the SOTA model.** Replacing MC-dropout with a deep ensemble of $M$=5 independently-trained Transolver members—using the per-cell standard deviation across members as $\sigma$—converts the near-constant band into a genuinely input-adaptive one while preserving coverage (Table 3). On the same 100/100 calibration/test split, the conformal quantile drops from $\approx 10^9$ to $q = 2.35$, the reliability ECE improves from $\approx 0.31$ to **0.074**, and coverage holds at 0.915 (target 0.90, consistent with the $\approx \pm 0.03$ binomial spread at $n_{\text{test}} = 100$). This is a single ensemble training run, not a 3-seed average; the calibration/test *split*, however, is not the fragile part—re-drawing it 20 times leaves coverage at 0.899±0.009 with $q = 2.26 \pm 0.05$ on pressure (split-resampling study, subsection 5.9), the published split sitting at the upper end of that spread. The ensemble mean is also more accurate than any single member ($\rho_{C_l} = 0.9995$, $\rho_{C_d} = 0.998$, $C_l/C_d$ relative force error 4.4%/5.7%; `results/uq_ensemble/uq_results.json`). The practical recipe follows: MC-dropout suffices when a constant-width certificate is acceptable, and a deep ensemble is what makes the band track the error on a near-deterministic SOTA surrogate.

**Corrector vs. ensemble: two points on a cost axis.** The ensemble mean (`mse_u/v/p` = 0.077/0.058/440) is in fact *more accurate* than the DEQ-corrected single backbone (0.116/0.079/471), so we do not claim the corrector is the best available accuracy mechanism. The two answer different questions on a cost axis: the deep ensemble is the better *accuracy* play when 5× the training (and $M\times$ the inference) is affordable, while the learned corrector is the better *cheap* play—one backbone plus a small corrector—demonstrating cheap learned correction on a competitive surrogate, not a leaderboard entry. (The ensemble's cost axis is itself measured: an $M$-subset study shows $M$=2 already supports the triage decisions while full band adaptivity needs $M\approx$4–5; see section 7.) The ensemble and the corrector are complementary: the ensemble supplies the adaptive $\sigma$ the conformal certificate needs, and either prediction can be wrapped by the same trust layer.

Table 4: In-distribution AirfRANS `full` ablation, 3 seeds, mean ± std (population std). Lower MSE is better; $\rho$ closer to 1 is better; resid↔err $\rho > 0$ supports the trust signal. Bold marks the best arm per column among the physics-loss arms; the physics-loss-free arm (starred where best overall) is reported separately because it removes the physics objective entirely. See Figure 3.

| arm | `mse_u` | `mse_v` | `mse_p` | surf. `mse_p` | $\rho_{C_l}$ | $\rho_{C_d}$ | resid↔err $\rho$ |
|---|---|---|---|---|---|---|---|
| backbone | 3.479±0.105 | **0.385±0.012** | **2444.8±120.7** | 548823±27946 | 0.9868±0.0005 | 0.895±0.013 | 0.397±0.003 |
| backbone (no phys. loss) | **1.995±0.042** | 0.323±0.008* | 1963.5±55.5* | 1123649±104822 | **0.9920±0.0002** | **0.945±0.008** | 0.605±0.016 |
| backbone + local corr. | 3.482±0.058 | 0.398±0.007 | 2469.4±71.0 | 541534±20453 | 0.9854±0.0017 | 0.888±0.012 | 0.389±0.031 |
| backbone + DEQ corr. | 3.457±0.811 | 0.880±0.064 | 3832.7±331.3 | **361681±12273** | 0.9856±0.0017 | 0.923±0.014 | **0.827±0.002** |

### 5.3 The grid backbone: the corrector trades accuracy for trust signal

This ablation uses the **weak grid backbone**; it establishes the backbone-dependence we report as a finding, and should be read against the SOTA-backbone result in subsection 5.2, where the same corrector *does* reduce field MSE.

**On this backbone the corrector trades accuracy.** The DEQ corrector is **flat** on volume `mse_u` (3.457 vs backbone 3.479; the per-seed deltas are sign-inconsistent, 2/3, so this is noise, not an improvement) and **worse** on `mse_v` (0.385 → 0.880, +129%, 3/3, non-overlapping distributions) and `mse_p` (2445 → 3833, +57%, 3/3). The feed-forward `local` corrector helps on **nothing**: it is at-or-worse than the backbone on every volume MSE channel and 3/3 worse on $\rho_{C_d}$. On this weak backbone the residual-conditioned correctors do not improve volume accuracy; the DEQ arm buys surface-pressure MSE (−34%, 3/3) and a slightly better drag ranking ($\rho_{C_d}$ 0.895 → 0.923, 3/3) at a measured cost to volume velocity and pressure. We do **not** claim the corrector improves accuracy in aggregate *on this backbone*—and this is precisely the contrast that makes correction quality backbone-dependent.

**The detector succeeds.** The same DEQ arm roughly **doubles** the residual–error Spearman correlation, 0.397 → 0.827 (3/3, std ≈ 0.002, Cohen's $d \approx 175$; effect sizes here and below use the population std, `ddof=0`, matching the tables). A low residual tracks low field error far more reliably with the corrector engaged. This is the detector half of the thesis, and it is the largest, cleanest effect in the table.

**A control that sharpens the thesis.** Removing the physics loss entirely (`backbone (no physics loss)`) **improves** `mse_u`, `mse_v`, `mse_p`, $\rho_{C_l}$, $\rho_{C_d}$, and the residual–error correlation (all 3/3), at the cost of **doubling** surface-pressure MSE (549k → 1124k, 3/3). The physics-loss-free backbone is in fact the **best force-ranking model in the table** ($\rho_{C_d}$ 0.945, $\rho_{C_l}$ 0.992)—better than any corrected model, with no correction loop at all. We therefore explicitly do **not** claim the loop delivers best-in-class force ranking. The physics objective (in the loss or the corrector) buys surface-pressure fidelity and a strong trust signal, not volume accuracy or force ranking *on this weak backbone*—the trust-signal role succeeds while the correction role does not, which is exactly the backbone-dependence the SOTA result in subsection 5.2 resolves.

**Version note for readers of the preprint.** The single-seed figures reported in earlier public versions of this preprint—$\rho_{C_l}$ rising 0.924 → 0.958 and surface MSE −25% under the corrector—do not replicate: across 3 seeds the backbone already sits at $\rho_{C_l} = 0.987$ and the DEQ corrector at 0.986, flat-to-slightly-worse and sign-inconsistent. Table 4 supersedes them.

### 5.4 Residuals detect under distribution shift

We evaluate each arm on the regime-disjoint `reynolds` and `aoa` splits (trained on the train range, tested on the held-out range—true extrapolation); Table 5 reports both.

**The trust signal is regime-invariant—the single strongest result.** On the `aoa` split the bare backbone's residual–error correlation **collapses to** 0.314, while the DEQ-corrected model holds it at 0.746 (0.314 → 0.746, 3/3, Cohen's $d \approx 8.9$, non-overlapping: every corrected seed ≥ 0.72, every backbone seed ≤ 0.38). On `reynolds` both are high (≈ 0.74–0.75). The corrected model thus keeps a regime-invariant

Table 5: Out-of-distribution AirfRANS ablation (regime-disjoint train→test), 3 seeds, mean ± std. The full four-arm table is in `results/full_research/ood/ablation_ood.md`. See Figure 4.

| task | arm | `mse_u` | `mse_v` | `mse_p` | surf. `mse_p` | $\rho_{C_l}$ | $\rho_{C_d}$ | resid↔err $\rho$ |
|---|---|---|---|---|---|---|---|---|
| reynolds | backbone | 16.218±0.746 | 1.187±0.149 | 7220.7±612.2 | 964641±61419 | 0.950±0.008 | 0.893±0.009 | 0.748±0.077 |
| reynolds | + DEQ | 5.300±1.544 | 1.208±0.255 | 6469.5±438.0 | 652281±45390 | 0.943±0.008 | 0.915±0.009 | 0.742±0.041 |
| aoa | backbone | 4.312±0.077 | 1.310±0.039 | 6042.8±242.5 | 2000592±55092 | 0.963±0.003 | 0.926±0.002 | 0.314±0.064 |
| aoa | + DEQ | 3.538±0.681 | 1.407±0.089 | 4620.6±170.2 | 1042711±78298 | 0.966±0.006 | 0.905±0.013 | 0.746±0.027 |

$\approx 0.74$ trust signal across both shifts, precisely where the one-shot model loses it. The residual is a reliable detector exactly where detection matters most.

**Two negatives the OOD ablation preserves.** The OOD ablation also shows the DEQ corrector reducing the in-distribution→OOD *gap* on volume velocity, volume pressure, and surface pressure (e.g. `reynolds` `mse_u` 16.2 → 5.3, `aoa` surf. `mse_p` 2.00M → 1.04M, both 3/3). We report this as an observation about the loop's behaviour, **not** as a competitiveness or generalisation guarantee, and we explicitly preserve two negatives. (i) **`mse_v` is an attenuated-regression artifact, not a gain:** the DEQ corrector's *absolute* OOD `mse_v` is worse than the backbone in every regime (0.880/1.208/1.407 vs 0.385/1.187/1.310); its in-dist→OOD gap only looks smaller because its in-distribution `mse_v` is already inflated. (ii) **Drag ranking is task-dependent:** $\rho_{C_d}$ improves on `reynolds` (0.893 → 0.915, 3/3) but **regresses on `aoa`** (0.926 → 0.905, worse on all 3 seeds). We do not claim the loop uniformly preserves force ranking under shift. Methodological caveat: the in-distribution reference is the `full` split (a different training range), so the gap mixes a train-set change with regime shift, and the `aoa` `mse_u` effect, while 3/3 directional, is seed-0-dominated.

**A second dataset: the trust signal generalizes beyond AirfRANS.** The strongest test of generality is a different dataset entirely. We evaluate the residual trust signal on **DeepCFD** (Ribeiro et al., 2020)—2-D laminar incompressible flow over bluff bodies (circles, squares, rhombi): a different geometry family *and* a different regime (laminar, versus AirfRANS's turbulent RANS). We train an FNO backbone (3 seeds) on an out-of-distribution split that holds out the largest bodies—forcing a range of per-case errors—and measure the same per-case residual–error Spearman correlation (interior continuity+momentum residual; `scripts/run_deepcfd.py`, `results/deepcfd/deepcfd_results.json`). The trust signal holds (Figure 9): **`residual_error_spearman`** $= 0.770 \pm 0.119$ (3 seeds), comparable to—indeed above—the AirfRANS Transolver value (0.611). The error spread is genuine (per-case rel-$L_2$ median 0.063, max 0.136), so the correlation is not degenerate; per-seed case-level bootstrap 95% CIs ($10^4$ resamples) are $[0.65, 0.77]$, $[0.58, 0.72]$ and $[0.91, 0.95]$, all far from zero (`results/control/bootstrap_spearman_ci.json`). A clean mechanistic reason it can be *sharper* than the turbulent case: laminar flow has $\nu_t = 0$, so the residual is the *exact* Navier–Stokes residual with no turbulence-closure ambiguity—a cleaner error indicator. This is direct evidence that the residual-as-trust-signal property is not specific to AirfRANS, its geometry family, or its turbulence model. (We report only the trust signal here; DeepCFD's force coefficients carry the same self-integrator caveat as subsection 5.2 and are not headlined.)

**Beyond regime shift: a geometry the model never saw (qualitative).** The splits above shift the flow regime *within* the airfoil family; we close with a qualitative probe of cross-*geometry* extrapolation. We run the airfoil-trained Transolver on a *cylinder*—a bluff body absent from AirfRANS—and ask not whether the prediction is good (it is not) but whether the residual—a data-free quantity evaluated against the governing PDE, needing no reference field (unavailable for a cylinder at these *Re*)—registers the geometry as out-of-distribution. It does, in the residual *magnitude*: the predicted cylinder field is uniformly less PDE-consistent than in-distribution airfoils with good predictions ($\approx 0.6\%$ rel-$L_2$). Normalising each case's residual by its freestream $U^2$ (the momentum-residual scale), the dimensionless far-field residual is $\approx 7\times$ higher on the cylinder (median $\approx 30\times$; Figure 10, `results/figures/cylinder_control_airfoil_ratio.json`). This gap holds in the *far field*, away from any wall, so it is a genuine geometry-OOD effect—neither a near-wall discretisation artifact nor a velocity-scale effect (it survives the $U^2$ normalisation). **Scope, with a controlled negative.** This is a qualitative illustration on one geometry family, and we report a control that *refutes* the tempting spatial reading: the residual's bright near-body *ring* is a generic no-slip-wall

Table 6: Iteration sensitivity sweep: `results/sensitivity/iters.csv`. Single sweep (one checkpoint, not seeded). See Figure 6.

| `n_iters` | `mse_u` | surf. `mse_p` | residual_norm | resid↔err $\rho$ |
|---|---|---|---|---|
| 0 | 3.924 | 541204 | 0.113 | 0.423 |
| 1 | 2.460 | 428560 | 0.336 | 0.644 |
| 3 | 2.287 | 336718 | 0.542 | 0.647 |
| 5 | 2.439 | 308381 | 0.594 | 0.675 |
| 10 | 2.570 | 300298 | 0.618 | 0.703 |
| 15 | 2.575 | 300664 | 0.620 | 0.710 |

feature, not an OOD localiser—in-distribution airfoils with good predictions show the *same* near-body-vs-far-field concentration ratio ($\approx 23$) as the cylinder ($\approx 18$), so spatial near-wall concentration cannot be read as "where the prediction failed." The out-of-distribution signal is therefore the field-wide residual *magnitude*, not its spatial ratio; and the residual remains a local-violation detector, not a global-correctness oracle. It complements, not replaces, the quantitative regime-shift results above.

### 5.5 The iteration sweep: cutting the error buys no residual reduction

The direct evidence that the residual is a poor *objective* is the descent experiment of subsection 5.7, which minimises it. This subsection reports a weaker, complementary observation—an iteration sweep on a single trained checkpoint (Table 6)—and the four limits that keep it from carrying the claim alone.

**Four limits of this sweep.** *(i)* The swept knob is the DEQ corrector's internal fixed-point cap (`results/sensitivity/iters.json`), and that corrector is trained by supervised regression toward ground truth; it never sees the residual as an objective. The sweep is therefore an anti-correlation along one trajectory, not a test of residual minimisation. *(ii)* The anti-correlation is carried by one channel. Across the sweep the residual rises while surface `mse_p` falls, and those two are strongly anti-correlated (rank correlation $-0.94$); `mse_u` has essentially no relationship with the residual ($-0.03$). `mse_u` falls only over iterations $0 \to 3$ and then *rises* alongside the residual ($2.287 \to 2.575$ against $0.542 \to 0.620$), so on three of the five intervals the two move in the *same* direction. Row 0 is moreover backbone-only, which makes $0 \to 1$ a corrector on/off contrast rather than an iteration step; restricted to $n_{\text{iters}} \geq 1$ the residual rises 84.7% while `mse_u` rises 4.7%—again the same direction. The $0 \to 3$ `mse_u` endpoints alone therefore overstate the effect. *(iii)* It is a single unseeded checkpoint ($n = 1$), and it reports `mse_u` and surface `mse_p` only, while Table 4 shows this corrector family inflating `mse_v` and volume `mse_p`. *(iv)* A seeded five-seed re-run on the Transolver+DEQ arm (`scripts/iters_sweep_seeded.py`, 24 cases) returns DIRECTION-ONLY against a bar declared in advance: the residual is higher at $k$=15 in 5/5 seeds but by only +1.5%, at 1.3× the across-seed standard deviation where we required 2×; it is monotone in 0/5 seeds; and paired per case it is higher in only 45/120 = 38%. The sweep therefore supports *decoupling*, not divergence, and we claim only that cutting field error along this path buys no residual reduction. None of these limits applies to subsection 5.7, which is why the claim now rests there.

As the loop iterates, the **PDE residual norm rises monotonically** ($0.11 \to 0.62$) while the **field error falls** (`mse_u` $3.92 \to 2.29$ by iter 3)—over that initial interval the two objectives move in *opposite* directions. Beyond iteration 3 they move together, as noted above, so the reading this table supports is that the residual is not a usable stopping signal along this path rather than that it is anti-correlated with error throughout. (Note this does not contradict the DEQ contraction of subsection 5.9: the contraction is in the correction variable $\delta$, which converges; the PDE residual of the *output field* is a different quantity, and it is the one that rises.)

**The acceptance gate is a throttle, not a filter—and it works.** The acceptance test is satisfiable by the identity (zero step), so its *definition* suggests it should admit almost nothing: a corrector that cannot reduce the residual is simply rejected. Measured, it does the opposite. Over 1200 gated steps (`scripts/measure_acceptance_gate.py`, `results/control/acceptance_gate.json`; 200 AirfRANS `full` test cases × 3 seeds, on both the deployed Transolver+DEQ system and the ensemble-mean path, with the accept rule, backtrack count and tolerance imported from the shipped `correction_loop` so the gate under test is the deployed gate), the test admits a step on 99.8% of deployed cases—the *full* correction on 50.7% of them—and refuses outright on one case in 600. The mechanism is backtracking: where the full step would raise the residual it is refused, but a halved step lowers it and is admitted (on the ensemble path the residual ratio falls from a median 1.027 at the full step to 0.975 at half). Crucially the admitted step is not empty progress: it lowers true rel-$L_2$ error on 89.3% of deployed cases, by a median **5.8**% (74.3% and 2.6% on the ensemble-mean path, where the correction has less headroom because the ensemble mean is already the more accurate field, subsection 5.2). The "certified self-correction" guarantee is therefore real *and* useful: a cheap, provably non-worsening gate that also happens to improve accuracy.

**The accuracy is damping, not the residual test.** The obvious control is an *ungated* fixed half step: take half the correction unconditionally, never consulting the residual. It improves true error on 95.8% of deployed cases by a median 6.2%, against the gate's 89.3% and 5.8%. The gate's accuracy advantage is therefore not there; what improves the field is damping the correction, and the residual test is not what earns that. What the gate uniquely buys is the **certificate**. The ungated half step raises the monitored residual on 1.0% of deployed cases and 8.8% on the ensemble path (`results/review/control3_fixed_step.json`), and the gate makes that outcome impossible by construction rather than improbable in practice. On the path a user actually runs the guarantee therefore binds on roughly one case in a hundred: it is a guarantee, not an accuracy claim, and that is what we claim for it.

This does not rescue the residual as an *objective*—the full step still raises the residual on half the deployed cases while error falls, and Table 6 shows the two decoupled under iteration. The gate and the objective are different mechanisms and earn separate verdicts. (Scope: a single gated step applied counterfactually to the deployed DEQ correction, which the shipped DEQ path bypasses by design; not a re-run of the multi-iteration feed-forward loop of Table 4, whose corrector checkpoints were not retained.) We additionally note that for the DEQ corrector the trust-gating and acceptance-test toggles are structural no-ops—the DEQ branch bypasses both—so the only live correction knob is the applied-delta step size (`results/sensitivity/toggles.json`).

### 5.6 The floor does not refine away—it grows

If the floor $r^\star = R_h(u^\star)$ were an artifact of the $128^2$ grid under-resolving the boundary layer, everything built on it would collapse to "you under-resolved your data". It is not. We rasterise the same AirfRANS truth at $128^2$, $256^2$ and $512^2$ and measure the floor on each (`scripts/floor_resolution_ladder.py`).

**Protocol, fixed before the measurement.** Three choices decide this experiment and all three were registered in the repository before it was run. *(i)* The scored region is fixed in **physical** units—fluid cells with sdf $\geq 1.5\,h_{128} = 0.0352$ chord—because the solid-adjacent ring the monitor zeroes is physically thinner on a finer grid, so scoring the default set would compare different regions at different levels and could manufacture a plateau by unmasking the steepest cells. *(ii)* The reading of the fitted exponent in $\|r^\star\| \sim Ch^p$ was fixed in advance: $p \geq 1$ with an extrapolation to zero is DECAY, $p < 0.5$ is PLATEAU. *(iii)* Levels whose spacing falls below the source cloud's own are excluded, since below that the raster differentiates a piecewise-linear reconstruction rather than sampling a solution.

**Result.** The floor rises monotonically as the grid is refined (Table 7): $0.0624 \to 0.0779 \to 0.1067$ while $h$ falls by 4×, rising in **21/24** cases, with $p = -0.387$—not merely inside the pre-registered PLATEAU band but the wrong *sign* for decay. (These are band-restricted values, scored on the physically fixed region of protocol *(i)*, and are therefore smaller than the whole-grid RMS floor of mean 0.192 quoted in section 6; the two score different regions and are not directly comparable.) Meanwhile $\|R_h(u_\infty)\| = 0$ exactly at every level, as it must: a spatially constant field has identically zero finite differences at any $h$. The gap the objective must be blamed for therefore *widens* with refinement.

**Four controls, because a rising floor is usually a bug.** *The operator is second order.* Method of manufactured solutions on the same monitor gives $p = 2.06$ in the interior band (fitted on the rung means;

Table 7: The residual floor under grid refinement, 24 AirfRANS test cases, scored on a region fixed in physical units. The floor grows; the uniform freestream stays at exactly zero. Fitted $p$ in $\|r^\star\| \sim Ch^p$; the pre-registered rule reads $p < 0.5$ as a plateau.

| level | $h$ | $\|r^\star\|$ | continuity | momentum | $\|R_h(u_\infty)\|$ |
|---|---|---|---|---|---|
| $128^2$ | 0.0234 | 0.0624 | 0.0477 | 0.0401 | 0 |
| $256^2$ | 0.0117 | 0.0779 | 0.0589 | 0.0510 | 0 |
| $512^2$ | 0.0059 | **0.1067** | 0.0791 | 0.0717 | 0 |
| fitted $p$ | | −0.387 | −0.365 | −0.419 | — |
| rose in | | 21/24 | 21/24 | 21/24 | — |

$2.02 \pm 0.05$ across the per-case fits), so the discretisation converges as claimed and the behaviour on real data is a statement about the data. The manufactured field is a potential flow, on which the monitored viscous term vanishes identically, so this control bounds truncation error and is *not* evidence that the operator is consistent (section 6). *Restoring the omitted closure term does nothing.* Adding $(\partial_j \nu_t)(\partial_j u_i + \partial_i u_j)$ back changes each rung's mean floor by −0.01% to +0.18% across five rungs, and no individual case by more than 1.5% (the per-case range is −1.06% to +1.45%, wider than the range of means and still two orders of magnitude too small to matter); the term itself is 3.5–4.8% of the floor. At $Re \approx 2 \times 10^6$ the entire viscous term is 1–4% of convection, so no part of it can carry a floor of this size—the floor lives in the convection/pressure block. *A higher-order interpolant does not reduce it.* Rasterising the same source cloud with cubic rather than linear interpolation *raises* the floor by 6–12%, and lowers it in only 2–4 of 16 cases; if linear interpolation's kinks set the floor, cubic would cut it. *The pipeline does not manufacture rising residuals.* Pushing a manufactured analytic solution through the identical rasteriser gives a residual that *falls* with refinement (fine/coarse 0.41–0.74) where the real truth's rises.

**What this leaves.** Two statements, which we keep separate. The inconsistency of the monitored operator gives the floor a nonzero **continuum limit**, so refinement alone cannot remove it (section 6). Its measured **size**, however, is set by the mismatch between our Cartesian stencil and the body-fitted discrete balance the AirfRANS reference actually solved—a mismatch common to every equation, which is why continuity and momentum rise at the same rate in the same cases. We report the closure term as a small, separately provable component and do not attribute the observed floor to it. A limitation of the ladder itself: at $512^2$ the raster spacing is only 1.27× the source cloud's far-field spacing, so that level is near the exclusion limit; the $128^2 \to 256^2$ leg, comfortably clear of it, already shows the rise.

### 5.7 Descending the residual directly

Testing the residual as an objective requires minimising it, which the corrector sweep of Table 6 does not do: that corrector is trained by supervised regression toward ground truth and never sees the residual as an objective. We minimise it directly here (`scripts/residual_descent.py`).

**Design.** We run 300 Adam steps on $J(y) = \frac{1}{2}\|R_h(y)\|^2$ using the differentiable residual, over 24 AirfRANS test cases, from two starting points. The first is the **exact ground truth**—no network is involved anywhere, so the standard reply that a surrogate is over-smoothed has nothing to attach to. The second is the truth plus a smooth displacement, which is where a surrogate's error actually lives. Steps are taken in units of each channel's own scale, since a uniform absolute step would destroy $\nu_t$ (order $10^{-3}$) long before the velocity (order 10) moved. As a control we descend the true field error from the same perturbed start; it must, and does, recover the truth.

**Result: the truth is not a fixed point.** Starting at $u^\star$, descent cuts the monitored residual by 84% in **24/24** cases and drives the field error from exactly zero to a median of 0.91—the range in which a trained backbone *starts*. This is Theorem 2(ii) as a measurement rather than an algebraic remark: the same descent applied to the *error* objective from the same point does not move at all, because $\nabla\|e\|^2 = 0$ there exactly,

Table 8: Armijo-line-searched gradient descent on the monitored objective $J = \frac{1}{2}\|R_h\|^2$, 500 steps, $n = 200$ AirfRANS `full` test cases, with the far-field and near-wall Dirichlet data pinned at ground truth. $J$ is monotone non-increasing by construction. rel-$L_2$ is the fluid-masked rel-$L_2$ of speed (the `measure_acceptance_gate` formula); $\Delta$ is the median per-case change. $\|R_h\|$ here is the whole-grid norm, so the truth's 0.191 is the same measurement as the fluid-cell 0.192 quoted elsewhere, rescaled by the geometric factor 0.995. `scripts/residual_descent_test.py`.

| start field | $\|R_h\|$ start→end | $J/J_0$ | $\Delta$ rel-$L_2$ | frac. worse | `mse_u` start→end |
|---|---|---|---|---|---|
| ground truth $u^\star$ | $0.191 \to 0.125$ | 0.398 | — $(0 \to 0.0069)$ | **200/200** | $0.000 \to 0.499$ |
| Transolver seed 0 | $0.218 \to 0.144$ | 0.406 | +76.5% | 0.965 | $0.119 \to 0.556$ |
| Transolver seed 1 | $0.218 \to 0.142$ | 0.388 | +73.9% | 0.940 | $0.120 \to 0.553$ |
| Transolver seed 2 | $0.217 \to 0.141$ | 0.390 | +86.9% | 0.980 | $0.103 \to 0.544$ |
| dropout-FNO | $0.286 \to 0.153$ | 0.238 | −18.7% | 0.250 | $0.898 \to 0.930$ |

while $\nabla J \neq 0$. The compact-stencil monitor we report falls in 24/24 cases too, so this is not the differentiable twin's wider Laplacian being gamed.

**What holds universally is iterate selection.** We do *not* claim residual descent always harms the field: from the perturbed start it *reduces* error in 18/24 cases, typically by 60%, before overshooting. What holds universally is a statement about **iterate selection**. The error-minimising and residual-minimising iterates never coincide—they differ in 24/24 cases on both arms—and the residual-selected iterate is strictly worse in **24/24**. From the perturbed start the error bottoms at step ≈62 and the residual keeps falling to step ≈192, by which point the error is 1.3× its best value on the path. A practitioner who stops where the residual says lands worse than a point already traversed. Descent also terminates at a residual of 0.028 while the exact truth sits at 0.110: the objective ranks a field with error 1.31 as four times better than the field with error zero.

**At $n = 200$, on the deployed backbone, and under gradient descent.** The preceding arms use 24 cases, an Adam optimiser and a synthetic displacement. We repeated the experiment with the framework's *monitored* operator (a differentiable replica verified against `Diagnostics.residual_norm` to $10^{-8}$ relative), under **Armijo-line-searched gradient descent**—so $J$ is monotone non-increasing and the step is chosen by the objective—on 200 AirfRANS test cases, starting from the *deployed* Transolver (three backbone seeds) and from the dropout-FNO of Table 6, with the far-field and near-wall Dirichlet data *pinned at ground truth* so the negative cannot be attributed to an unconstrained residual (Table 8; `scripts/residual_descent_test.py`). Unpinning those boundaries makes the negative stronger, not weaker—the error then rises on 100/99.5/100% of cases across the three seeds—so the pinned arm is the conservative one and is what we report (`results/residual_descent/descent_*_free_uvp_armijo.json`). From the truth, $J$ falls to 39.8% of $J_0$ while the field error rises on **200/200** cases and never returns to its starting value at any of 500 steps, ending at rel-$L_2$ = 0.0069—1.57× the *entire* error of the deployed Transolver—at a measured exchange rate $\Delta\mathrm{err}/\Delta\|R_h\| = -0.140$. From the deployed Transolver the residual falls 34% while the error rises on **96.5/94.0/98.0**% of cases (seeds 0/1/2; median +76.5/+73.9/+86.9%; Wilcoxon $p < 10^{-30}$), against the supervised corrector's −9/−21/−25% on the same backbone. The verdict is unchanged under three step rules—Armijo, fixed $\eta$ swept over 12 decades, and Adam—and in every one the outcome is monotone in the *achieved* $J$: the harder the objective is minimised, the worse the field.

**Where descent helps, and why that does not rescue the objective.** The dropout-FNO row is a genuine exception: there descent *lowers* the error, on 75% of cases. The boundary is measurable in the theorem's own variable $\rho = \|R_h(\hat{u})\|/\|r^\star\|$, since the truth arm supplies $\|r^\star\|$ for the same case. Over 200 cases the fraction on which residual descent lowers rel-$L_2$ rises monotonically with $\rho$: for the dropout-FNO 0.40 at $\rho \in [1, 1.5)$, 0.97 at $[1.5, 2)$ and 1.00 above 2; for the deployed Transolver 0.01, 0.28 and 0.25. The deployed backbone has median $\rho$ = 1.13–1.14 with 88% of cases in the lowest bin: it is already inside the regime where descent almost never helps. ($\rho$ is a strong trend, not a complete predictor—at matched $\rho \in [1.5, 2)$ the two backbones differ—so we claim the ordering, not a law.) Even in the favourable regime the objective is not usable: the gain is ≈1/35 of the supervised corrector's, and there is **no stopping rule**—$J$ falls monotonically on 100% of cases while the error minimum is interior on 78.5%, so driving $J$ lower with Adam cuts the gain from −15.3% to −6.7%.

Two further readings we record because they went against our own expectation. The error does not rise monotonically—these are Adam steps on a non-convex objective, not gradient flow—so the claim is about endpoints, not paths. And the iterate moves *away* from the uniform freestream rather than toward it: descent does not travel to the spurious global minimum of leg (i), it finds a nearby different low-residual field. The damage does not require reaching $u_\infty$, which makes it more relevant to a short practical polish, not less.

### 5.8 The boundary is the operator, not the problem class

Residual-driven correction is not doomed in general, and it is important to say exactly where it works. Lei et al. (2026) take a surrogate prediction as an initial guess for Newton–Krylov correction on *steady* CFD and lower the median residual $L_2$ ratio by over seven orders of magnitude *while* substantially reducing field and aerodynamic error—residual down and error down, on steady problems as here. Huang & Perdikaris (2025) report a similar success on *transient* rollouts, which matters for the boundary we are about to draw: the two successes differ in problem class but share the property that does the work.

That is not a counterexample to our result; it locates it. In those works the residual being driven is the *solver's own* discrete operator, the one whose root the reference solution is. Its floor at the truth is zero by construction, so minimising it and minimising the error are the same problem. Our monitor is the affordable surrogate-side one: a different discretisation, on a different mesh, from the generator of the labels. Its floor at the truth is *not* zero, does not vanish under refinement (subsection 5.6), and its minimiser is displaced from $u^\star$ (section 6).

We therefore draw the boundary at **operator consistency**. The obvious alternatives—steady versus transient, resolved versus under-resolved—are falsified by Lei et al. (2026) being steady and successful. The operator boundary is not: it predicts success exactly where the monitored residual is the generator's own, and failure exactly where an affordable proxy replaces it.

The observation that a PDE residual need not vanish at the ground truth is not ours alone—Zhang et al. (2026) report it for shock physics and conclude that a physics-informed loss "cannot be formulated using the absolute PDE residual alone; instead, it must be defined relative to the residual present in the ground-truth data". Their remedy requires $R_h(u^\star)$, which a deployment-time monitor by definition does not have.

We also note the genre this paper belongs to. McGreivy & Hakim (2024) show that weak baselines and reporting biases have produced systematic overoptimism in machine learning for fluid PDEs; the corrective in both cases is a measurement rather than a method, and the controls we report above—an ungated baseline against our gate, a physics-free score against our physics one—are exactly the comparisons that work argues are routinely omitted.

Our contribution relative to Zhang et al. (2026) is to establish that the floor has a nonzero continuum limit for an inconsistent operator, to measure that it grows under refinement rather than merely persisting, and to show what the resulting misplaced minimum does to a correction path.

### 5.9 The conformal trust layer: coverage and contraction

**Contraction (H5).** Iterating the learned DEQ operator $\delta_{k+1} = T_\theta(\delta_k)$ from a random initialisation on 24 real AirfRANS cases, the measured per-step ratio $\|\delta_{k+1} - \delta^*\|/\|\delta_k - \delta^*\|$ (in the geometric regime, before the $\sim 10^{-7}$ solve floor) has **median 0.78** and maximum 0.86—strictly below the design bound $\kappa = 0.9$ and the falsification threshold 1—and reaches a relative distance of $10^{-5}$ to the fixed point in a median of 37 steps (`results/certificates/h5_contraction.json`). The corrector contracts as designed; this is a property of the *operator*, separate from whether minimising the output's PDE residual helps the field (it does not, subsection 5.5).

**In-distribution coverage (H4).** With per-cell MC-dropout $\sigma$ (16 passes) and a 100/100 calibration/test split of the AirfRANS test set, split-conformal calibration at $\alpha = 0.1$ attains per-channel coverage of **0.911 ($u$), 0.928 ($v$), 0.942 ($p$)**—all in the $[0.85, 0.95]$ band and conservatively above the 0.90 target, as the $\geq 1 - \alpha$ guarantee requires. The conformal multipliers ($q = 0.77, 1.30, 1.75$) both tighten an over-dispersed and widen an under-dispersed raw $\sigma$, so the calibration is doing real work. Coverage tracks $1 - \alpha$ across a sweep of $\alpha$ (Figure 8). The reliability diagram is exact at the target level but over-covers at lower nominal levels (ECE $u/v/p = 0.064/0.093/0.130$), the signature of a heavy-tailed $|\text{error}|/\sigma$ ratio; we therefore claim coverage at the chosen $\alpha$, **not** full distributional calibration (`results/certificates/h4_coverage.json`).

**The certificate is calibrated on the deployed, corrected field—not the raw backbone.** NeuroForge ships the DEQ-corrected field, so the coverage guarantee must hold for *that* output; split-conformal calibration, however, applies to whatever predictor it scores, and the raw-backbone certificate does **not** transfer unchanged through the corrector. We verify this on the dropout-FNO backbone whose DEQ corrector materially changes the field (mean $|\text{corrected} - \text{raw}| \approx 5$), through the *same* `cases_to_error_sigma` path as H4 but with a corrected predict function (`scripts/verify_conformal_corrected_field.py`; 15/15 calibration/test). The clean isolation is a *within-run frozen-$q$ contrast*—same fitted $q$, same input-conditional $\sigma$, same cases, only the raw vs. corrected error map—and it shows the direction of the coverage change tracking the direction of the error change: where the corrector *enlarges* the error the raw band drops below the 0.90 target ($v$ $0.896 \to 0.819$, $p$ $0.864 \to 0.777$), and where it *shrinks* the error the raw band over-covers and stays conservative ($u$ $0.856 \to 0.888$). This is the coverage analogue of detector $\neq$ fixer (subsection 5.5): the DEQ operator contracts in its *own* variable $\delta$ (H5, median ratio 0.78), but the Banach bound governs the correction, not $|\text{error}|/\sigma$, so it cannot imply coverage survival—and on $v/p$ residual contraction coincides with error *growth*, which is exactly what breaks the raw band. Re-fitting $q$ on the corrected calibration field—at no change to $\sigma$, which is input-conditional and identical for both fields—restores coverage toward target where it had collapsed ($v$ $0.819 \to 0.894$, $p$ $0.777 \to 0.834$) and sharpens it where it was loose ($u$ $0.888 \to 0.874$). We therefore calibrate the shipped certificate on the corrected output; split-conformal validity requires only exchangeable corrected-field scores, not that $\sigma$ be the corrected field's own uncertainty. These estimates are directional at $n = 15$ (per-case coverage std $\approx 0.15$–$0.23$, so $\approx \pm 0.05$ standard error), but the same pattern—raw-$q$ under-covers $v/p$ on the corrected field, refitting restores it—reproduces on an independent 20/20 split (`results/certificates/probe_conformal_after_deq.json`); the residual $p$ under-coverage is the small-sample heavy-tail effect already noted at $n = 100$. On the deployed SOTA Transolver backbone we now verify this *directly* rather than by extrapolation: through the same conformal path, with $\sigma$ the frozen five-member deep-ensemble std and the scored field replaced by the DEQ-corrected output, at the production 100/100 split and across all three corrector seeds (`scripts/run_w2_conformal_corrected.py`, `results/uq_ensemble/w2_conformal_corrected.json`; a faithfulness gate reproduces the published backbone $p$-certificate—$q$=2.35, coverage 0.915—exactly). Re-fitting $q$ on the corrected calibration field gives a valid certificate (coverage $u/v/p = 0.88/0.91/0.91$ vs the 0.90 target), and because the corrected field's error is lower at frozen $\sigma$, that accuracy propagates into the certificate as proportionally tighter intervals at the same coverage ($q$: $v$ $2.58 \to 2.34$, $p$ $2.35 \to 2.10$ on every seed). Applying the *backbone's* own $q$ to the corrected field—the direct "remains-conservative" test—over-covers on $v$ and $p$ ($\approx 0.93$), as the field-error reduction (subsection 5.2) predicts, while $u$ holds at $\approx 0.88$, just below target: this tracks the backbone's own heavy-tailed $u$ band (0.881) and is inherited from it, not introduced by the corrector. We therefore ship the certificate refit on the corrected output; recalibration matters most for correctors that trade field error for residual reduction (the dropout-FNO above), and is benign-to-beneficial for the deployed Transolver, whose bands it sharpens.

**What the floor costs the certificate: valid, but wide.** The certificates above are stated on field-error targets. The quantity an engineer actually wants bounded is a force coefficient, and conformalising the monitored residual against $|\Delta C_D|$ shows exactly what the floor does to a certificate that remains formally valid. Over 400 random half-splits at a 0.90 target on the deployed backbone (`results/review/functional_audit_gate_followup.json`), coverage comes out at 0.891–0.895 across three seeds—so the guarantee holds, and the floor does *not* invalidate split conformal, which needs only exchangeability and never supposed the score vanished at the truth. What the floor destroys is tightness. The median bound is **5.6–6.8**$\times$ the median drag error it certifies, because the median truth-to-prediction residual ratio is 0.864: about 86% of a typical prediction's monitored score is already present at zero error, a fraction set by the operator rather than by the model, and a nonconformity score that is 86% constant cannot separate cases finely. subsection 5.6 closes the obvious escape—the floor grows under refinement, so a finer grid widens this certificate rather than narrowing it. This is the precise sense in which the audit can certify and yet not certify usefully, and it is why we describe the residual's certification role as priced rather than absent.

**Split-resampling robustness (20 splits).** Because the certificates above are computed on one 100/100 calibration/test draw, we re-draw the split 20 times (`scripts/run_multisplit_conformal.py`, `results/uq_ensemble/multisplit_conformal.json`; same code path, frozen ensemble $\sigma$, $q$ refit per split). Coverage is tightly centred on the 0.90 target in *every* arm and channel (0.895–0.902, split-to-split std $\leq$ 0.014); the published seed-0 pressure coverage of 0.9155 sits at the upper end of its distribution ($0.899 \pm 0.009$). The corrected-field sharpening is robust, not a split artifact: on pressure, $q$ drops from $2.26 \pm 0.05$ (backbone) to 2.02–2.07 for all three corrector seeds, and on $v$ from $2.55 \pm 0.05$ to 2.28–2.31; on the heavy-tailed $u$ channel the effect is mixed (one seed sharpens, two widen slightly), consistent with the $u$-neutral verdict above.

**Out-of-distribution coverage (empirical, not a guarantee).** Calibrating $q$ on the in-distribution `full` split and evaluating on the OOD splits, coverage stays in the target band: $u$ 0.90/0.91/0.91, $v$ 0.91/0.94/0.92, $p$ 0.92/0.94/0.90 (full / reynolds / aoa) vs the 0.90 target (`results/sensitivity/ood_coverage.json`, Figure 7). **This is an empirical observation, not a guarantee.** The conformal guarantee assumes exchangeable calibration and test draws, which fails under shift; coverage nonetheless holds because the MC-dropout $\sigma$ inflates appropriately on shifted inputs, so the same $q$ still brackets the (larger) errors. We report this as a desirable empirical property and explicitly **do not** assert distribution-free coverage out-of-distribution.

**From correlation to decisions: selective prediction.** A rank correlation of 0.6 can sound modest, so we also measure what the trust signal is *for*: deciding which predictions to reject (`scripts/run_selective_prediction.py`, `results/selective/selective_prediction.json`; Figure 2). On the 200 AirfRANS test cases (ensemble-mean arm), the residual detects worst-decile-error cases with AUROC 0.871; the ensemble $\sigma$ alone gives 0.894; and the rank-fused residual+$\sigma$ score reaches **AUROC** 0.905 (AUPRC 0.681 against a 0.10 prevalence baseline). The same residual score transfers: AUROC 0.89–0.91 on the three DEQ-corrected arms and 0.83–0.98 on the three DeepCFD OOD seeds. In risk–coverage terms, rejecting the 10% least-trusted cases by the fused score reduces the retained mean relative-$L_2$ error from 0.0040 to 0.0035—recovering $\approx 91\%$ of the error reduction an *oracle* that rejects by true error could achieve at that budget (the residual alone recovers $\approx 67\%$; random rejection recovers none). Case-level bootstrap 95% CIs ($10^4$ resamples; `results/control/bootstrap_spearman_ci.json`) put the ensemble-arm residual correlation at 0.610 $[0.506, 0.698]$ and the fused score at 0.703 $[0.615, 0.773]$; every arm's CI excludes zero by a wide margin. The trust signal is thus not merely correlated with error; it supports near-oracle triage at deployment-relevant rejection budgets.

### 5.10 Where the physics earns its place: drag

On field error, the physics does not. A *physics-free* score—the ensemble $\sigma$, which needs no operator, no mesh and no PDE—is nominally ahead of the residual (0.894 against 0.871), and a paired case-level bootstrap cannot separate them ($\Delta$AUROC +0.023, 95% CI $[-0.035, +0.083]$). The gain that survives is the *fusion*'s: the AUROC 0.905 of subsection 5.9 belongs to the fused score, not to the residual alone.

There is one place the physics wins outright, and it is the place an engineer cares about. Ranking cases by error in the *drag coefficient* rather than in the field, the residual reaches AUROC **0.952** against 0.871 on field error, and here it beats the physics-free score decisively ($\Delta$AUROC +0.055 to +0.088, bootstrap CIs excluding zero). The residual degrades gracefully onto the engineering quantity where the physics-free score does not.

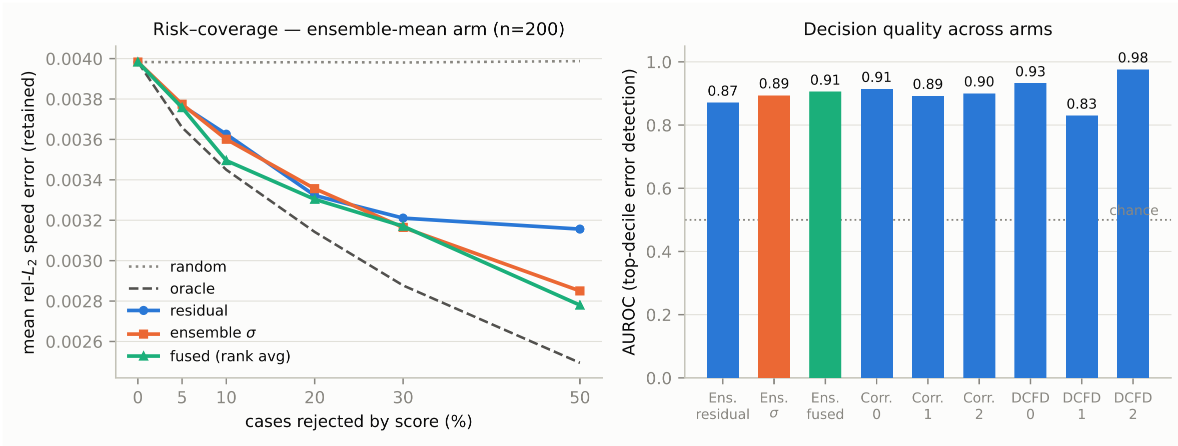


Figure 2: **Selective prediction with the trust signal.** *Left:* risk–coverage curves on the AirfRANS test set (ensemble-mean arm): mean retained relative-$L_2$ error after rejecting the least-trusted fraction of cases by physics residual, ensemble $\sigma$, or their rank fusion, against the oracle (reject by true error) and random rejection. *Right:* AUROC for detecting worst-decile-error cases across arms (AirfRANS ensemble-mean, three corrected seeds, three DeepCFD OOD seeds); chance = 0.5.

**The detector is not a difficulty proxy.** Cases whose geometry rasterises badly could be both high-$\|r^\star\|$ and hard to predict, which would make the correlation a restatement of case difficulty. Conditioning on each case's *own* floor—the ground-truth field's residual, an oracle quantity—leaves a partial rank correlation of 0.561 against a raw 0.610, so 92% of the association survives and the drop's CI includes zero. The deployable residual also significantly outranks that oracle difficulty variable itself, which a pure difficulty proxy could not do.

The audit that produces this score is essentially free at the point of use: a median 1.13 ms per case on a single CPU thread ($n = 100$, subsection 5.11; `results/control/inference_cost.json`), or 1.65 ms as a conservative bound from an independent 200-case re-measurement taken under CPU contention (`scripts/measure_audit_cost.py`, `results/control/audit_cost.json`). Either way the certificate costs $\sim 10^{-6}$ of the ∼25-minute, 16-core RANS solve that produced each AirfRANS case (Bonnet et al., 2022).

The trust layer depends only on a model's predictions and $\sigma$, so it is backbone-robust; subsection 5.2 demonstrates it on a SOTA Transolver backbone, coverage and band adaptivity included (Table 3).

### 5.11 Computational cost and complexity

A trust layer is only deployable if auditing costs materially less than the prediction it audits. We therefore time every stage of the deployed pipeline on the AirfRANS `full` test set at resolution 128 (`scripts/measure_inference_cost.py`, `results/control/inference_cost.json`; 100 cases, median of two timed repeats after an untimed warm-up, CUDA synchronised around each repeat, on one RTX 4070 Ti with the package's single-thread BLAS cap in force). Table 9 reports the result.

**The certificate is free; the ensemble is not.** The audit costs 1.13 ms against a 3822 ms solve—one part in 3 400, or 0.03%. End to end, a case costs 5.24 s (encode + solve + audit) against the ∼1500 s that produced its label, a ∼286× reduction, and the certificate is invisible inside that. This is the quantitative form of the paper's claim that the surrogate can audit itself: the trust score, the trust map and the conformal band are obtained for three hundredths of a percent of the computation they vouch for, and they require no ground truth.

The deep-ensemble spread does not share that property. Measured directly by loading five independent backbones, $M$=5 costs 4.86× a single backbone—linear in $M$, as expected, and confirmed rather than assumed—which is 2.1 deployed solves of *extra* compute. The two halves of the fused score therefore sit

Table 9: **Per-case wall-clock cost of the deployed pipeline** ($128^2$ grid, 100 AirfRANS `full` test cases, medians). *Where* is the device each stage actually runs on. The audit is the entire certificate: residual maps, trust map, case-level score and conformal band. The classical anchor is the cost of generating one AirfRANS case, as reported by its authors (≈ 25 min on 16 CPU cores); it is quoted for scale, not as a matched-hardware benchmark.

| stage | where | median | p95 | share of a deployed solve |
|---|---|---|---|---|
| encode (geometry → input planes) | CPU, 1 thread | 1413 ms | 1620 ms | preprocessing |
| backbone forward + rasterise | GPU | 2080 ms | 3537 ms | 54.4% |
| correction (Neural Residual Iteration) | GPU | 1747 ms | 2950 ms | 45.7% |
| **deployed solve** (backbone + correction) | GPU | **3822 ms** | 6400 ms | 100% |
| **audit** (residuals + trust + score + band) | CPU, 1 thread | **1.13 ms** | 2.49 ms | **0.03**% |
| deep ensemble, $M$=5 (for the fused score) | GPU | 10 114 ms | 10 546 ms | 265% |
| classical steady-RANS solve (Bonnet et al., 2022) | 16 CPU cores | ∼1 500 s | — | ∼39 000% |

Table 10: Matched-budget baseline on AirfRANS `full` (`n_train` = 800, 80 epochs, identical rasterisation and `evaluate_cases` scoring), 3 seeds. See Figure 5.

| model | `n_params` | `mse_u` | `mse_v` | `mse_p` | surf. `mse_p` | $\rho_{C_l}$ | $\rho_{C_d}$ |
|---|---|---|---|---|---|---|---|
| Transolver (baseline) | 7.35M | 0.120±0.005 | 0.088±0.013 | 628.5±29 | 9110±500 | 0.9992±0.0002 | 0.9963±0.0012 |
| our backbone (Table 4) | — | 3.479 | 0.385 | 2444.8 | 548823 | 0.987 | 0.895 |
| our backbone + DEQ | — | 3.457 | 0.880 | 3832.7 | 361681 | 0.986 | 0.923 |

at opposite ends of the cost scale, and this sharpens the selective-prediction result of subsection 5.9 into a deployment choice rather than a single operating point: the residual alone recovers ≈ 67% of the oracle's achievable error reduction at essentially zero marginal cost, while fusing it with the ensemble spread reaches ≈ 91% but triples the compute budget per case. A practitioner who cannot afford five backbones still gets two thirds of the benefit for nothing; one who can, buys the last third at a known, measured price.

**Scaling.** The audit's operation count is $O(N)$ in the number of grid cells $N = n_y n_x$: every term is a fixed-width finite-difference stencil, a mask, or a reduction. Empirically, over a 16× range in $N$ ($64^2$ to $256^2$, fields resampled purely to vary array extent) the audit rises only 6.3×, a log–log slope of 0.64 ($R^2 = 0.99$)—sub-linear because per-call fixed costs (allocation, Python dispatch) still dominate at these sizes. The asymptotic rate is the $O(N)$ of the operation count; the practical consequence is that the audit's share of the pipeline *falls* as grids grow, since the backbone and the corrector both carry super-linear cost in resolution. Across the test fleet the backbone's cost showed no measurable dependence on the native cloud size, but that range is only 1.21× wide (163 590–197 146 points) and cloud size explains 3% of the timing variance ($R^2 = 0.03$); we therefore report the observation and decline to fit a scaling law to it.

**What these numbers are not.** The backbone stage times the Transolver forward pass *and* the rasterisation of its point-cloud output onto the grid; the latter is single-threaded NumPy under the package's BLAS cap, so the GPU figures are conservative upper bounds on the model itself rather than a tuned inference benchmark. The encode stage is likewise unoptimised CPU preprocessing and would be cached in any real design loop, where the same geometry is queried at many operating points. The classical anchor comes from the AirfRANS authors on 16 CPU cores (Bonnet et al., 2022); ours is one consumer GPU plus one CPU thread, so the ∼286× figure indicates the order of the gap, not a controlled speed-up measurement. None of these caveats touch the ratio the paper actually depends on—audit against solve—because both sides of it were measured in the same run, on the same cases, under the same thread caps.

### 5.12 Matched-budget baseline

Transolver, a SOTA point-cloud physics-attention transformer, trained on the same data with the same budget and scored identically, is far ahead of our grid backbone: relative to the bare backbone, roughly

Table 11: Published AirfRANS `full`-task baselines from Bonnet et al. (2022), as field context only. Volume MSE on standardized fields ($\times 10^{-2}$); surface $\bar{p}$ ($\times 10^{-1}$); force relative errors (%); $\rho$ = Spearman rank correlation vs. *official* force labels. NeuroForge is *not* a comparable row (physical-unit MSE, self-integrated forces; subsection 5.2). The near-zero/negative $\rho_{C_d}$ is the benchmark's known drag-ranking difficulty.

| method | $\bar{u}_x$ | $\bar{p}$ | surf. $\bar{p}$ | $C_L$ err | $C_D$ err | $\rho_{C_l}$ / $\rho_{C_d}$ |
|---|---|---|---|---|---|---|
| MLP | 0.95 | 0.74 | 1.13 | 0.77 | 4.29 | 0.913 / −0.117 |
| GraphSAGE | 0.83 | 0.66 | 0.66 | 0.52 | 4.05 | 0.965 / −0.303 |
| PointNet | 3.50 | 1.15 | 0.93 | 0.74 | 14.64 | 0.938 / −0.022 |
| Graph U-Net | 1.52 | 0.66 | 0.39 | 0.49 | 10.39 | 0.967 / −0.138 |

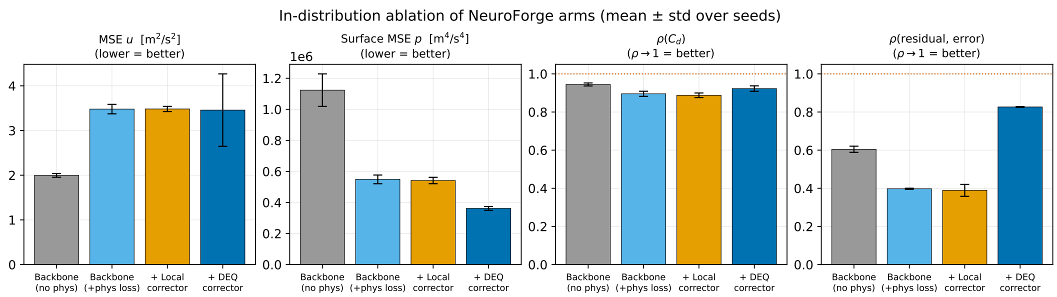


Figure 3: In-distribution ablation (Table 4): the DEQ corrector lifts the trust signal ($0.40 \to 0.83$) and surface fidelity while regressing volume `mse_v`/`mse_p`.

29× **on `mse_u`**, 4× **on `mse_v`** (≈ 10× against the volume-regressed DEQ arm), 4× **on `mse_p`**, and ∼ 60× **on surface pressure** (≈ 40× against the DEQ arm, whose surface MSE is lower), with near-perfect force ranking. Our grid backbone is not a competitive surrogate; we make **no competitiveness claim** for it. The decisive next step follows in subsection 5.2: we put the trust layer and the DEQ corrector *on* this Transolver backbone, where the residual is a calibrated detector ($\rho = 0.611$) and the corrector reduces field MSE by 8–25%. The large grid-backbone gap is itself informative: even a much weaker backbone yields a residual that is a calibrated, shift-robust detector of its own errors.

**Note on the two Transolver runs.** The Transolver numbers here (Table 10: `mse_u` 0.120±0.005, `mse_v` 0.088±0.013, `mse_p` 628.5±29) and the v2 backbone-alone numbers (Table 2, 5 seeds: 0.133±0.017, 0.096±0.006, 644±63) are the *same Transolver architecture at the same budget* (7.35M params, 80 epochs), trained in two separate runs; the differences are within the reported per-seed spread. The v2 run is the one we use as the NeuroForge backbone in subsection 5.2.

**Positioning against the published AirfRANS field.** Our trust layer and corrector run on a Transolver backbone—a published SOTA architecture (Wu et al., 2024)—so competitiveness comes from that architectural choice, not from re-benchmarking our own metric values. We deliberately do *not* place a NeuroForge row in a cross-method comparison: AirfRANS results are reported in mutually-incompatible conventions (the AirfRANS-paper baselines use standardized-field MSE and force coefficients against official OpenFOAM labels; Transolver-lineage papers report relative-$L_2$; our pipeline uses physical-unit MSE and self-integrated forces, subsection 5.2), so any shared cell would mislead. Table 11 reproduces the canonical AirfRANS-paper baselines as field context. Note the benchmark's signature difficulty—drag ranking $\rho_{C_d}$ is near zero or negative for every baseline, the headline finding of Bonnet et al. (2022)—which is also why we read our own $\rho_{C_d}$ as internal self-consistency rather than a solved drag-ranking claim.

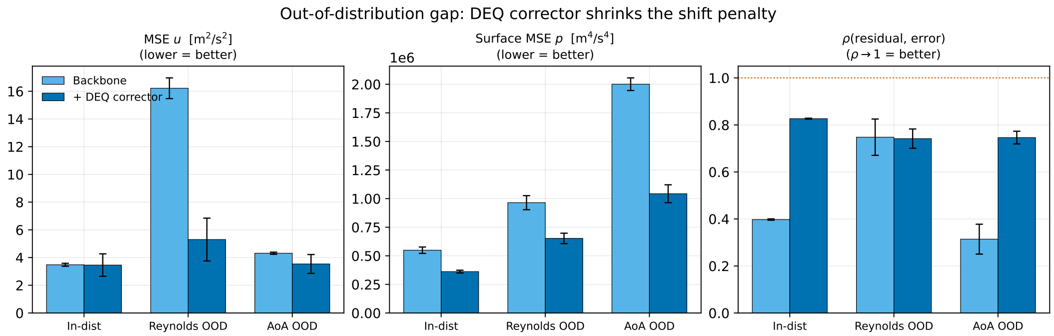


Figure 4: OOD gap and the regime-invariant trust signal (`aoa` 0.31 → 0.75); the bare backbone's signal collapses, the corrected model's holds.

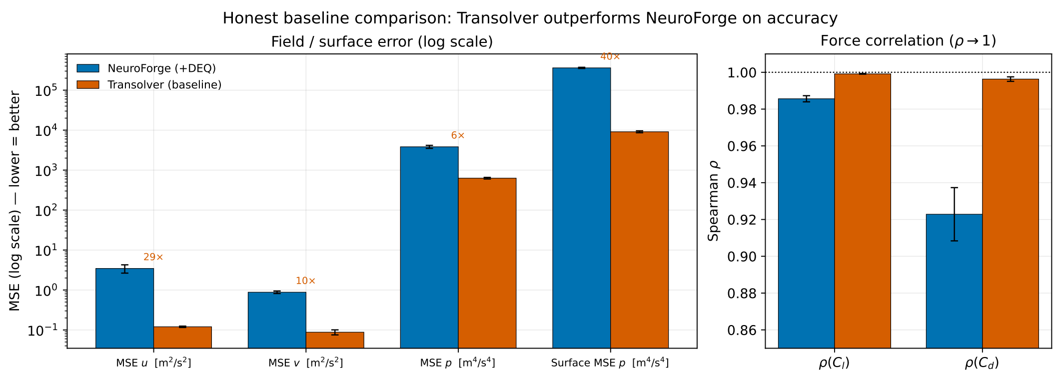


Figure 5: Matched-budget baseline: the grid backbone trails Transolver $\sim$ 4–60$\times$ across channels (no competitiveness claim).

### 5.13 Figures

### 5.14 Reading against the pre-registered hypotheses

- **H1 (the corrector improves accuracy).** *Backbone-dependent.* On the **weak grid backbone** it is *not supported as pre-registered*: the feed-forward corrector helps on nothing and the DEQ corrector regresses volume `mse_v`/`mse_p` (3/3) while flat on `mse_u`. On the **SOTA Transolver backbone** it *is supported*: the DEQ corrector reduces all three volume MSE channels on all 5 seeds (`mse_u` $-8\%$, `mse_v` $-21\%$, `mse_p` $-25\%$; subsection 5.2, Table 2) with force ranking held. Correction quality is therefore backbone-dependent, and the supervised corrector works where the backbone is strong.

- **H2 (the residual is a valid trust signal).** *Supported, robustly, in- and out-of-distribution.* resid↔err $\rho > 0$ for every arm/seed/split; DEQ lifts it to 0.83 in-dist and makes it regime-invariant ($\approx 0.74$) OOD, rescuing the `aoa` collapse 0.31 → 0.75. The registered condition is met with room to spare. Two qualifiers belong with it and are stated in full in subsection 5.10: on field error a physics-free ensemble $\sigma$ matches the residual ($\Delta$AUROC +0.023, CI $[-0.035, +0.083]$), and the physics wins outright only on drag (AUROC 0.952).

- **H3 (DEQ $\geq$ feed-forward corrector).** *Holds on the design axis only.* DEQ beats the feed-forward corrector on surface MSE, $\rho_{C_d}$, and the trust signal (3/3), but is worse on volume `mse_v`/`mse_p`; the two trade off.

- **H4 (conformal coverage).** *Supported in-distribution* (0.91/0.93/0.94 at $\alpha = 0.1$; SOTA backbone 0.902, deep-ensemble band adaptive at ECE 0.074, Table 3); *empirical only OOD* (subsection 5.9).

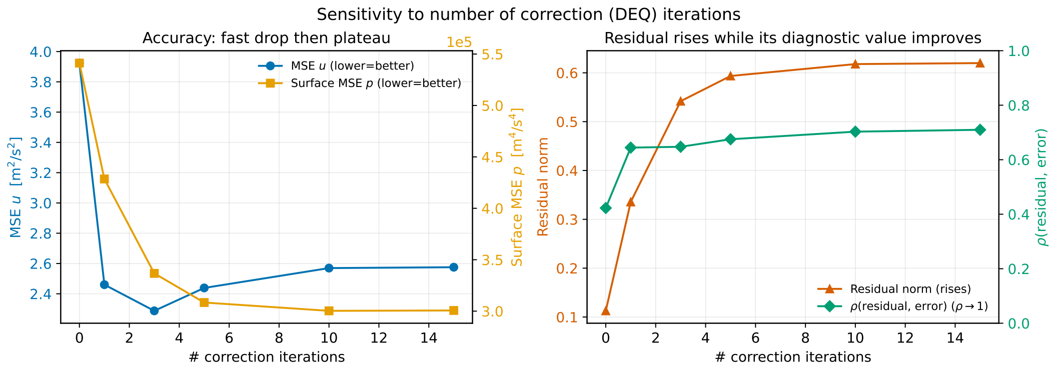


Figure 6: The iteration sweep (Table 6, one unseeded checkpoint). Over iterations $0 \to 3$ the PDE residual rises while `mse_u` falls; beyond iteration 3 the two move together. A seeded five-seed re-run returns DIRECTION-ONLY, so this figure illustrates *decoupling* between the residual and the error, not divergence. The claim itself rests on subsection 5.7.

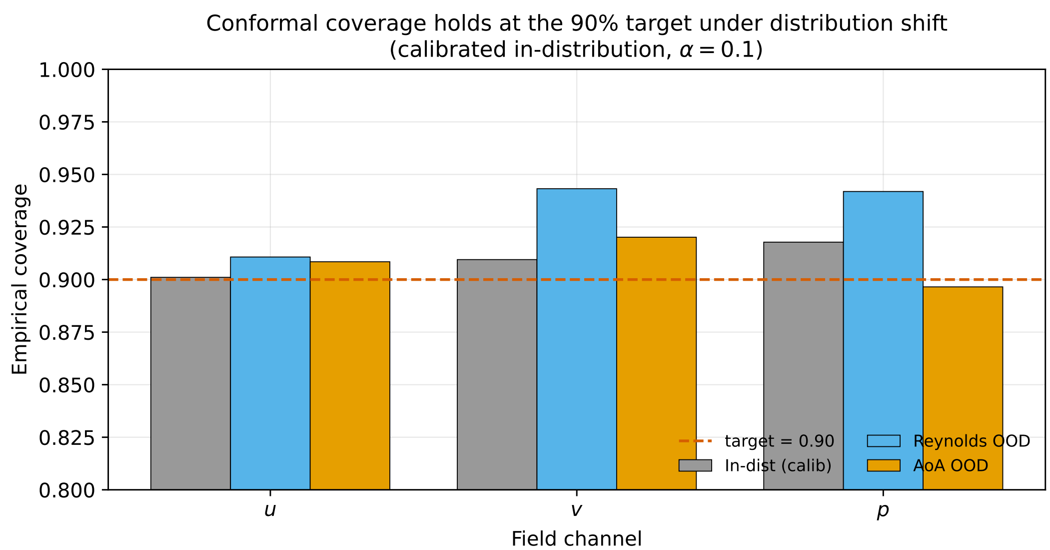


Figure 7: Conformal coverage stays in-band on OOD splits (empirical, via $\sigma$ inflation).

Coverage was never in doubt; *width* is what the floor costs. Conformalising the residual against $|\Delta C_D|$ stays valid (0.891–0.895 at a 0.90 target) at a median bound 5.6–6.8× the drag error it certifies (subsection 5.9).

- **H5 (DEQ contraction).** *Supported* (measured factor $0.78 < \kappa = 0.9 < 1$).
- **Not pre-registered: the residual as a correction *objective*.** The registered set predates subsection 5.7. We record the outcome here for completeness: descending $J = \frac{1}{2}\|R_h\|^2$ raises the field error on 200/200 cases from the truth and on 94–98% from the deployed backbone, so had it been registered it would be *falsified*.

Every number comes from the committed, reproducible harness (`benchmarks/ablation.py`, `scripts/run_certificates.py`, `results/sensitivity/`).

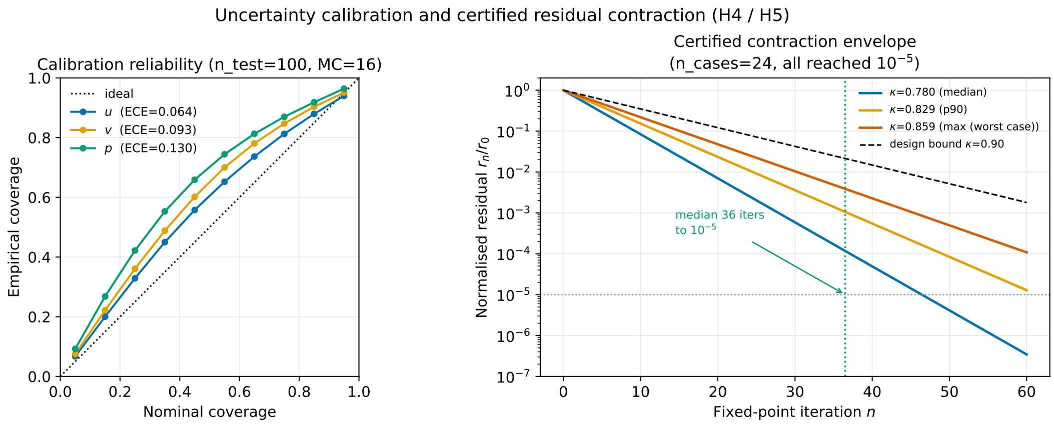


Figure 8: In-distribution reliability / coverage-vs-$\alpha$ and the measured DEQ contraction ($0.78 < 1$).

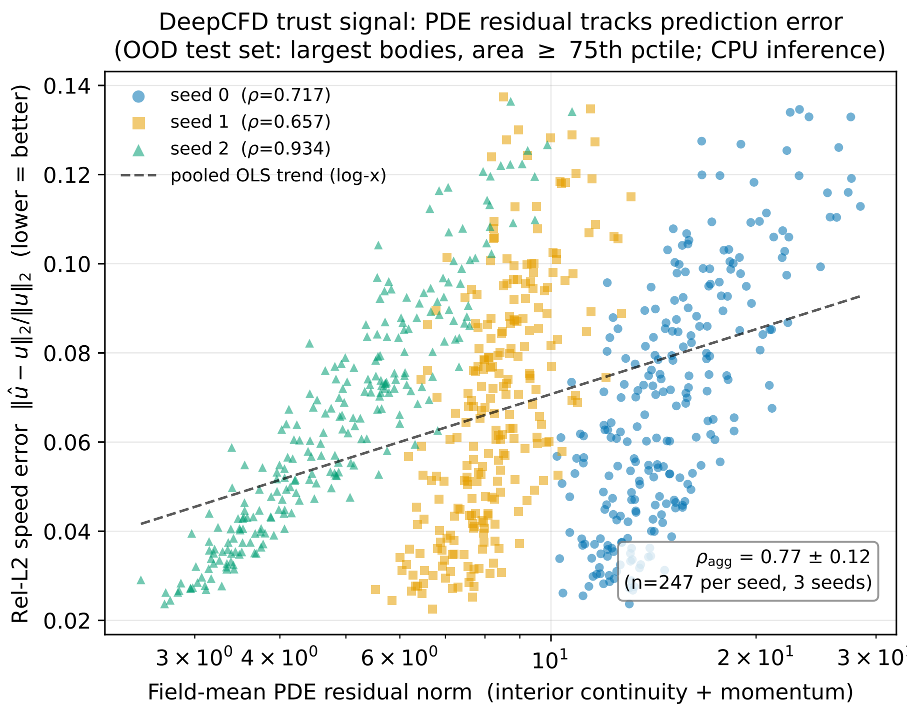


Figure 9: The trust signal generalizes to a second dataset (DeepCFD, 2-D laminar flow over bluff bodies; OOD held-out-largest-body split, 3 seeds). Per-case field-mean PDE residual norm (interior continuity+momentum) vs. rel-$L_2$ speed error: a low residual tracks a low error, as on AirfRANS. Per-case Spearman $\rho = 0.77 \pm 0.12$ (per seed 0.72/0.66/0.93; the trend holds *within* each seed—the colour-coded clusters span different residual ranges by initialisation). The residual is a reference-free error proxy on a geometry family and flow regime absent from AirfRANS.

## 6 The residual floor: what an operator-inconsistent monitor can and cannot do

The iteration sweep of subsection 5.5 (Table 6) shows a dissociation: under iteration the monitored physics residual *rises* ($0.113 \to 0.618$) while the field error *falls* ($3.92 \to 2.29$, before rising again to 2.58 at the largest caps). That is an anti-correlation along one trajectory of one checkpoint, not a causal claim; the causal experiment—descending $J = \frac{1}{2}\|R_h\|^2$ directly, with no network in the loop—is subsection 5.7. This

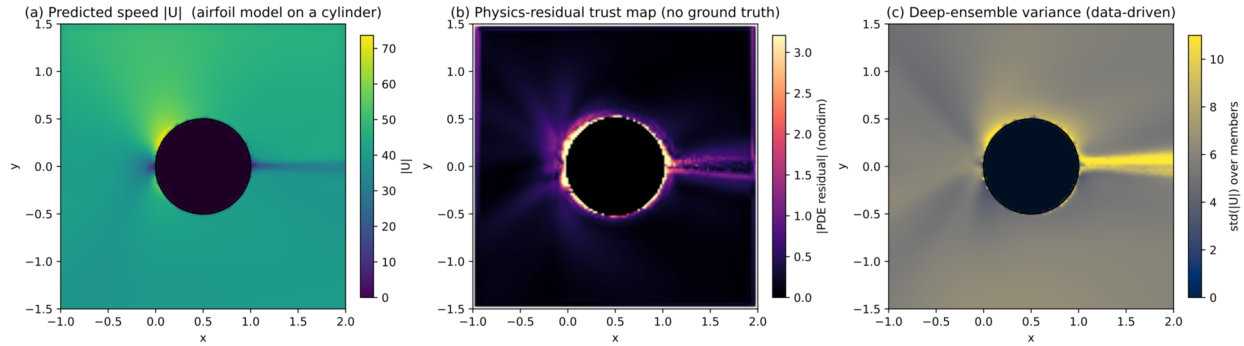


Figure 10: Cross-geometry OOD (qualitative): an airfoil-trained model on a cylinder. (a) predicted speed (visibly broken—the flow is not arrested at the body); (b) the physics-residual map (no ground truth used), elevated field-wide; its bright near-body ring is a generic no-slip-wall feature common to all bodies (a matched in-distribution-airfoil control shows the same near/far ratio), *not* an OOD localiser; (c) deep-ensemble variance (the uncorrected five members, the source of $\sigma$), for comparison. The OOD signal is the field-wide residual *magnitude*—$\approx 7\times$ higher than in-distribution airfoils in the far field, away from any wall (subsection 5.4)—not its spatial concentration.

section supplies the mechanism common to both, and it is not ill-conditioning. It is that the monitored residual is *self-consistent at a field that is not the truth*: the operator a deployed surrogate can afford to evaluate is not the operator whose root its training labels are, so the truth carries a floor and the objective's minimiser sits away from $u^\star$.

The section is organised around the paper's three verbs. **Certify: only loosely**, and this is the leg the theory settles—Theorem 1 shows the floor survives the continuum limit, and Theorem 2 shows the truth is neither a minimiser nor a stationary point, so no *threshold* on the residual norm implies a bound on the error. A calibrated bound is still available by conformalising the score against held-out data, and we report one; what the floor destroys is its tightness, since 86% of a typical prediction's score is present at zero error and the resulting certificate is 5.6–6.8× the quantity being certified (subsection 5.9). **Descend: no**, by Theorem 2(iii)–(iv), measured in subsection 5.7. **Rank: yes**—nothing here contradicts that, and we are explicit below about why a floor that defeats certification leaves between-case triage intact, because the same monitor flags worst-decile drag error at AUROC 0.952 (subsection 5.10).

### 6.1 Setup: the monitored operator, stated exactly

Let $\Omega \subset \mathbb{R}^2$ be the crop domain itself (a $3c \times 3c$ square about a unit chord here), $\Omega_h$ the uniform Cartesian grid on it, $h$ its spacing, $M \subset \Omega_h$ the fluid cells, $W \subset M$ the *wall ring* (fluid cells with a 4-neighbour in the solid), and $A := M \setminus W$ the **active set**. We write $\Omega_f \subset \Omega$ for the open fluid region, so $|\Omega_f| < |\Omega|$. The state is $u = (u, v, p, \nu_t)$. The monitored operator $R_h$ evaluates, on physical (denormalised) fields with $\nu_{\text{eff}} = \nu + \nu_t$ pointwise,

$$\begin{aligned} r_c &= D_x u + D_y v, \\ r_x &= u D_x u + v D_y u + D_x p - \nu_{\text{eff}} \Delta_h u, \\ r_y &= u D_x v + v D_y v + D_y p - \nu_{\text{eff}} \Delta_h v, \end{aligned} \tag{7}$$

with $D_x, D_y$ second-order central in the interior and first-order one-sided at the two edges of each axis, and $\Delta_h$ the compact three-point second difference per axis. All three are set to zero on the solid and on $W$, then non-dimensionalised by $U/\ell$ (continuity) and $U^2/\ell$ (momentum), with $U$ the freestream speed and $\ell$ the reference length (we reserve $L$ for the Jacobian below). The scalar we write $\|R_h(\cdot)\|$ in this section is the deployed one, `Diagnostics.residual_norm`: the root-mean-square over the **whole grid** of $\sqrt{r_c^2 + r_x^2 + r_y^2}$ after that non-dimensionalisation, with **no boundary term**. We name it because the paper also reports band-restricted norms over a region fixed in physical units when it studies refinement (subsection 5.6); the two are different scoring regions and their numbers are not interchangeable.

Write $J(u) = \frac{1}{2}\|R_h(u)\|^2$, let $u^\star$ be the discrete ground-truth field, define the *residual floor* $r^\star := R_h(u^\star)$, and for a prediction $\hat{u}$ write $e := \hat{u} - u^\star$ with

$$R_h(\hat{u}) = r^\star + L\,e + o(\|e\|), \qquad L := DR_h(u^\star), \tag{8}$$

where—here and throughout—$L$ denotes the Jacobian restricted to the fluid, $L : \mathbb{R}^{4|M|} \to \mathbb{R}^{3|A|}$.

### 6.2 The floor does not vanish in the continuum limit

The objection this theorem answers is the one a numerical analyst raises first: *a discrete operator applied to a non-solution leaves a truncation error; refine the grid.* It does not apply, because $R_h$ is not a low-order discretisation of the RANS equations. It is a *consistent* discretisation of a *different* system. Recall that $R_h$ is consistent with a PDE system $F(u) = 0$ if $R_h(u|_{\Omega_h}) \to 0$ pointwise as $h \to 0$ for every sufficiently smooth solution $u$ of $F$.

**Theorem 1** (Consistency floor: (H2) is a theorem, not an assumption)**.** *Let $(u^\star, p^\star, \nu_t)$ solve the steady incompressible RANS system on the open fluid region $\Omega_f \subset \Omega$ of subsection 6.1, with a* linear (Boussinesq) eddy-viscosity closure*, in which the modelled deviatoric Reynolds stress is $-2\nu_t S$ with $S = \frac{1}{2}(\nabla u + \nabla u^\top)$, and with the isotropic part absorbed into the modified pressure:*

$$u_j \partial_j u_i + \partial_i P - \partial_j[\nu_{\text{eff}}(\partial_j u_i + \partial_i u_j)] = 0, \qquad \partial_j u_j = 0. \tag{9}$$

*Let $K \subset\subset \Omega_f$ be compact with $u^\star \in C^4(K)$ and $\nu_t \in C^1(K)$. Then:*

(a) (The omitted term, in closed form.) *At every grid node in $K$, for the* dimensional *residuals of (7) evaluated on $u^\star|_{\Omega_h}$ (the per-equation scalings are reinstated in (c)),*

$$\begin{aligned} r_c = O(h^2), \qquad r_i \;&=\; T_i + O(h^2), \\ T_i := (\partial_j \nu_t)(\partial_j u_i &+ \partial_i u_j) = 2(S\nabla\nu_t)_i, \end{aligned} \tag{10}$$

*and in two dimensions, where incompressibility makes $S$ symmetric and* traceless*,*

$$\begin{aligned} |T| \;=\; 2\lambda_S\,|\nabla\nu_t| \;&=\; \sqrt{2}\,\|S\|_F\,|\nabla\nu_t| \quad \text{exactly}, \\ \text{with } \lambda_S = &\sqrt{(\partial_x u)^2 + \tfrac{1}{4}(\partial_y u + \partial_x v)^2} \end{aligned} \tag{11}$$

*the strain-rate eigenvalue (distinct from the no-slip weight $\lambda$ swept later in this section).*

(b) (Non-degeneracy.) *$T(x) = 0$ if and only if $S(x) = 0$ or $\nabla\nu_t(x) = 0$. In particular no orthogonality hypothesis is required.*

(c) (The floor survives refinement.) *If $\left|\{x \in K : S(x) \neq 0 \text{ and } \nabla\nu_t(x) \neq 0\}\right| > 0$, then*

$$\liminf_{h\to 0} \left\|R_h(u^\star|_{\Omega_h})\right\| \;\geq\; \left(\tfrac{|K|}{|\Omega|}\right)^{1/2} \frac{\ell}{U^2} \left(\frac{1}{|K|}\int_K |T|^2\,dx\right)^{1/2} \;>\; 0. \tag{12}$$

*where $|\Omega|$ is the area of the* whole crop*, since $\|\cdot\|$ averages over the whole grid; the factor $(|K|/|\Omega|)^{1/2} \leq 1$ is the price of scoring a norm over $\Omega$ while bounding it only on $K$. Hence $R_h$ is* ***inconsistent*** *with (9): hypothesis* (H2) *of Theorem 2 is a consequence, not an assumption, and no grid refinement removes the floor.*

*Proof.* (a) Expand the stress divergence in (9) using $\partial_j[\nu_{\text{eff}}(\partial_j u_i + \partial_i u_j)] = \nu_{\text{eff}}\nabla^2 u_i + \nu_{\text{eff}}\partial_i(\partial_j u_j) + (\partial_j\nu_{\text{eff}})(\partial_j u_i + \partial_i u_j)$. The middle term vanishes identically by incompressibility of the *exact* solution, which is where this theorem lives; the discrete divergence of the rasterised field is a different quantity and is discussed separately below. Also $\nabla\nu_{\text{eff}} = \nabla\nu_t$ because the laminar viscosity is constant. Substituting into (9) gives $u_j\partial_j u_i + \partial_i P - \nu_{\text{eff}}\nabla^2 u_i = T_i$, whose left-hand side is exactly the continuum form of the monitored $r_i$ in (7). Central first differences are $O(h^2)$ on $C^3$ fields and the compact second difference is $O(h^2)$ on

$C^4$ fields, uniformly on the compact $K$; the products with the bounded coefficients $u, v, \nu_{\text{eff}}$ preserve that order. Continuity is a consistent discretisation of $\nabla\cdot u^\star = 0$, so $r_c = O(h^2)$ with no $O(1)$ term. For (11): in two dimensions incompressibility gives $\partial_y v = -\partial_x u$, so $S = \left(\begin{smallmatrix} a & b \\ b & -a \end{smallmatrix}\right)$ with $a = \partial_x u$, $b = \frac{1}{2}(\partial_y u + \partial_x v)$. Then $\det S = -(a^2 + b^2)$ and the eigenvalues are $\pm\lambda_S$, $\lambda_S = \sqrt{a^2 + b^2}$. As $S$ is symmetric its singular values are $|\pm\lambda_S| = \lambda_S$, both equal, so $|Sx| = \lambda_S|x|$ for *every* $x$; apply this to $x = \nabla\nu_t$ and use $\|S\|_F = \sqrt{2}\lambda_S$. (b) Immediate from $|T| = 2\lambda_S|\nabla\nu_t|$: a product of non-negative scalars vanishes iff a factor does, and $\lambda_S = 0$ iff $S = 0$. (c) On $K$ the non-dimensionalised integrand converges uniformly to $(\ell/U^2)^2|T|^2$ by (a), so the Riemann sums of the mean square over $K \cap \Omega_h$ converge to the stated integral. Writing $n_\Omega$ for the number of grid cells and $n_K$ for those in $K$, and using that every discarded cell contributes non-negatively,

$$\mathrm{MS}_\Omega = \frac{1}{n_\Omega}\sum_{\Omega_h}|R_h|^2 \;\geq\; \frac{1}{n_\Omega}\sum_{K\cap\Omega_h}|R_h|^2 \;=\; \frac{n_K}{n_\Omega}\,\mathrm{MS}_K,$$

and $n_K/n_\Omega \to |K|/|\Omega|$ as $h \to 0$ since the grid is uniform on the crop. Taking square roots and then $\liminf$ gives (12). Positivity follows from (b) and the positive-measure hypothesis. □

**Where the hypothesis fails, and why that is reassuring.** Theorem 1(b) says the floor degenerates exactly when $S = 0$ or $\nabla\nu_t = 0$. Both hold for the uniform freestream, which is precisely the field that Theorem 2(i) shows has an exactly zero monitored residual. The theorem therefore predicts its own exception, and the exception is the spurious minimum. The non-degeneracy hypothesis is also weak and checkable rather than cosmetic: it asks only that the flow deform and the eddy viscosity vary somewhere in $K$, which is the definition of a turbulent boundary layer or wake.

**What this proves, and—emphatically—what it does not.** It proves that the floor is bounded away from zero as $h \to 0$: the "refine the grid" objection is closed *in principle*. It does **not** explain the floor's measured size, and we do not claim it does. Measured on the same fields with the same stencils, the omitted term $T$ is $0.0016 \to 0.0048$ across the refinement ladder against a floor of $0.0459 \to 0.1074$—about 3.5–4.8%—and *repairing the full symmetric stress divergence moves the floor by under* 0.1% ($0.10737 \to 0.10740$ at $512^2$; these four figures come from the 16-case, five-rung study scored on the 0.1-chord band, so they are not interchangeable with the 24-case figures of subsection 5.6, which use a different exclusion radius; the two agree on the trend and on this fraction, which is what the argument uses; `results/certificates/floor_resolution_decomposition.json`). What sets the size is the mismatch between our Cartesian stencil and the body-fitted discrete balance the reference solver actually closed, and the evidence for that is subsection 5.6: continuity and momentum rise at the same rate in the same cases, which no single-term story predicts—a closure omission moves momentum alone, and lost divergence-freedom moves continuity alone. The two results are complementary and neither substitutes for the other: **the theorem closes refinement in principle, the ladder closes it in practice**, and the theorem's term is a small, separately provable component that guarantees the floor cannot be refined away even if every other source were eliminated. The algebraic identity of (10) is verified numerically as a committed unit test, converging at order 1.93.

**Why our own method-of-manufactured-solutions control cannot see this.** The refinement study gates on a manufactured potential flow and measures order $p = 2.06$ in the interior band, which confirms the stencils are second-order and might be read as confirming the operator is consistent. It cannot: for a potential flow $u = \nabla\phi$ one has $\nabla^2 u \equiv 0$, so the monitored viscous term vanishes identically and the manufactured field is an exact solution of the monitored operator *for any* $\nu_t$. The same artifact records this directly—multiplying $\nu_t$ by 50 changes the monitored norm by $6.7 \times 10^{-6}$ relative. An MMS gate built on a field that annihilates the very term at issue measures truncation and is structurally blind to inconsistency. The gate certifies the discretisation; it cannot certify the operator, and we state the blind spot alongside it.

### 6.3 The truth is not a minimiser, and residual descent leaves it

**Theorem 2** (Residual floor: misplaced minimum, displaced minimiser, local divergence)**.** *Assume* (H1) $R_h$ *is the monitored discrete operator of* (7) *(continuity + momentum, no-slip omitted); and* (H2) *the floor is* *nonzero, $r^\star := R_h(u^\star) \neq 0$—which Theorem 1 supplies, and subsection 5.6 measures. Write $L := DR_h(u^\star)$, let $\sigma_{\max}$ be its largest singular value and $P := P_{\operatorname{range} L}$. No injectivity is assumed; $L$ has a large kernel (Proposition 3). Then:*

(i) (Misplaced minimum, exact, and at every $h$.) *The uniform-freestream field $u_\infty$ satisfies $R_h(u_\infty) = 0$ exactly, so $\min_u J = 0$, whereas $J(u^\star) = \frac{1}{2}\|r^\star\|^2 > 0$. Hence $u^\star$ is **not** a minimiser of $J$. This cannot be an artifact of the rasterisation or of the grid: a spatially constant field has identically zero finite differences at* every *$h$ and for* every *stencil in (7), one-sided edges included.*

(ii) (Misplaced stationarity.) *$\nabla J(u^\star) = L^\top r^\star$, so $u^\star$ is stationary for $J$ iff $L^\top r^\star = 0$; if $r^\star \notin \ker L^\top$, gradient flow on $J$ leaves $u^\star$.*

(iii) (Displacement of the minimiser, bounded below.) *The minimisers of the linearised objective $\frac{1}{2}\|r^\star + Le\|^2$ form the affine set $e_\infty + \ker L$ with $e_\infty = -L^+ r^\star$ ($L^+$ the Moore–Penrose pseudoinverse). **Every** point $e$ of that set satisfies*

$$\|e\| \;\geq\; \|e_\infty\| \;\geq\; \frac{\|P\,r^\star\|}{\sigma_{\max}}, \tag{13}$$

*which is strictly positive whenever $r^\star$ has any component in* $\operatorname{range} L$.

(iv) (Local divergence direction.) *Under residual-gradient flow $\dot{e} = -\nabla J(\hat{u})$, $\frac{d}{dt}\frac{1}{2}\|e\|^2\big|_0 = -\|Le\|^2 - (Le)\cdot r^\star$, which is positive on the open cone $(Le)\cdot r^\star < -\|Le\|^2$. The cone is non-empty whenever $Pr^\star \neq 0$, and is reached by $Le = -\alpha\, Pr^\star/\|Pr^\star\|$ for $0 < \alpha < \|Pr^\star\|$.*

*Proof.* (i) Every stencil in (7) annihilates a constant—central differences term-by-term, and the one-sided edge stencils $(f_1 - f_0)/h$ and $(f_0 - 2f_1 + f_2)/h^2$ by inspection—so $R_h(u_\infty) = 0$ identically, giving $J(u_\infty) = 0 = \min J$ since $J \geq 0$. By (H2), $J(u^\star) > 0$. (ii) $\nabla J(u) = DR_h(u)^\top R_h(u)$; evaluate at $u^\star$ and use (8). (iii) $\frac{1}{2}\|r^\star + Le\|^2$ is convex, and its minimiser set is the solution set of the normal equations $L^\top Le = -L^\top r^\star$, namely $e_\infty + \ker L$ with $e_\infty = -L^+ r^\star$; no injectivity is needed. Since $L^+$ maps into $\operatorname{range}(L^\top) = (\ker L)^\perp$, any minimiser is $e = e_\infty + k$ with $k \in \ker L$ orthogonal to $e_\infty$, so $\|e\|^2 = \|e_\infty\|^2 + \|k\|^2 \geq \|e_\infty\|^2$: the bound holds uniformly over the minimiser set, not merely at the minimum-norm tie-break. For the second inequality, $LL^+ = P$ gives $Le_\infty = -Pr^\star$, whence $\|Pr^\star\| = \|Le_\infty\| \leq \sigma_{\max}\|e_\infty\|$. (iv) To first order $\dot{e}|_0 = -L^\top(r^\star + Le)$, so $\frac{d}{dt}\frac{1}{2}\|e\|^2|_0 = -e^\top L^\top(r^\star + Le) = -\|Le\|^2 - (Le)\cdot r^\star$, positive iff $(Le)\cdot r^\star < -\|Le\|^2$. Since $Le$ is confined to $\operatorname{range} L$, only the *projected* floor is reachable, which gives the constant $\|Pr^\star\|$ and not $\|r^\star\|$. ☐

**Scope of legs (iii) and (iv).** Legs (iii) and (iv) of Theorem 2 are statements about the *linearised* objective at $u^\star$ and about gradient flow; the deployed corrector is supervised toward truth and the acceptance gate uses the residual as a one-bit veto, so neither leg describes the deployed loop. They describe residual descent—which this paper does run, from the exact truth and with no network anywhere in the loop, and which behaves as (ii) and (iv) predict: $\nabla J(u^\star) \neq 0$ while $\nabla\|e\|^2(u^\star) = 0$ exactly, and 300 steps cut the residual by 84% in 24/24 cases while taking the field error from zero to a median of 0.91 (subsection 5.7). The nonlinear, non-asymptotic version of (iii)–(iv) is that measurement; the theorem is the local mechanism. We also do not report a number for (13): $\sigma_{\max}$ and $\|Pr^\star\|$ are not measured here, so (iii) is a strictly-positive-displacement result and we do not describe it as a quantified detection limit.

### 6.4 The kernel: what the monitor cannot see at all

The undetectable subspace is not a matter of a few isolated null modes. It is large, and its size is provable by counting.

**Proposition 3** (Structure of the undetectable subspace). *With $M$, $W$, $A = M \setminus W$ and $L : \mathbb{R}^{4|M|} \to \mathbb{R}^{3|A|}$ as above:*

(K1) (Counting.) $\dim\ker L \;\geq\; 4|M| - 3|A| \;=\; |M| + 3|W| \;\geq\; |M|$. *At least **one quarter** of the fluid state space is invisible to the monitor at first order, for the structural reason that $R_h$ imposes three* *equations per cell on four unknowns per cell, plus three further dimensions for each wall-ring cell whose residuals are zeroed outright.*

(K2) (Solid degrees of freedom, exactly and nonlinearly.) *If the body is at least three cells from the outer border of the crop, then every degree of freedom on a solid cell lies in* $\ker R_h$ exactly*—not merely in* $\ker L$*:* $R_h$ *is unchanged by* any *perturbation supported on the solid, of any size.*

(K3) (The $\nu_t$ block is diagonal, in closed form, exactly.) $R_h$ *is* affine *in* $\nu_t$*, which enters only as the pointwise multiplier* $-\nu_{\text{eff}}\Delta_h u$*. Hence for any perturbation* $\delta$ *supported on the* $\nu_t$ *channel with* $\nu + \nu_t + \delta \geq 0$*,*

$$R_h(u^\star + \delta) - R_h(u^\star) = \left(0,\ -\delta \odot \Delta_h u^\star,\ -\delta \odot \Delta_h v^\star\right)\Big|_A \quad \textit{exactly,} \tag{14}$$

*so* $\|R_h(u^\star + \delta) - R_h(u^\star)\|^2 = \sum_{k\in A} c_k^2 \delta_k^2$ *with* $c_k = \frac{\ell}{U^2}\sqrt{(\Delta_h u_k^\star)^2 + (\Delta_h v_k^\star)^2}$*. Consequently, for* $\Omega_\varepsilon := \{k \in A : c_k \leq \varepsilon\}$ *and any* $\delta$ *supported in* $\Omega_\varepsilon$*, the monitor moves by at most* $\varepsilon\|\delta\|$*: the sublevel set* $\{u : \|R_h(u)\| \leq t\}$ *contains a ball of radius* $(t - \|r^\star\|)/\varepsilon$ *inside a coordinate subspace of dimension* $|\Omega_\varepsilon|$*, for every* $t > \|r^\star\|$*.*

(K4) (A Nyquist pressure mode the monitor barely registers.) *Let* $p_\varepsilon(i,j) = \varepsilon(-1)^{i+j}$*. Then* $D_x p_\varepsilon = D_y p_\varepsilon = 0$ *at every* interior *node, so an* $O(1)$ *checkerboard oscillation in the pressure changes the monitored residual at* no *active cell except those on the outer border ring of the crop—an* $O(1/N)$ *fraction of the active set—and would be* exactly *invisible were that ring excluded, as the wall ring already is.*

*Proof.* (K1) Rank–nullity, with $\operatorname{rank} L \leq 3|A|$ and $|A| = |M| - |W|$; the final inequality is $|M| + 3|W| \geq |M| = \frac{1}{4}\cdot 4|M|$. (K2) Every interior stencil in (7) reads only a cell's four neighbours ($D_x$ reaches $(i \pm 1, j)$ and $D_y$ reaches $(i, j \pm 1)$; the compact $\Delta_h$ reaches both pairs). By definition of $W$, no active cell has a solid 4-neighbour, so no active cell's stencil reads a solid value. The one-sided edge stencils reach two cells inward, which is why the three-cell separation from the outer border is required; it holds for the $3c \times 3c$ crop about a unit chord used here. The statement is nonlinear because $R_h$ simply never evaluates those entries. (K3) $\nu_{\text{eff}} = \nu + \nu_t$ multiplies $\Delta_h u$ pointwise and appears nowhere else in (7), so $R_h$ is affine in $\nu_t$ with the stated diagonal linear part; the clip $\max(\nu_{\text{eff}}, 0)$ is inactive under the stated hypothesis, which holds on the reference data since $\nu_t \geq 0$ there. (K4) $(-1)^{(i+1)+j} = (-1)^{(i-1)+j}$, so the central difference $(p_{i+1,j} - p_{i-1,j})/2h$ vanishes identically, and likewise in $y$; only the first-order one-sided stencils at the four borders return a nonzero value, $2\varepsilon/h$. ☐

**Reading the kernel.** (K3) has a physical statement: *eddy viscosity is unobservable from a momentum residual wherever there is nothing to diffuse.* In the freestream $\Delta_h u^\star \approx 0$, so $c_k \approx 0$ and $\nu_t$ may be set arbitrarily there without moving the monitor at all. This is why we never claim the residual certifies the $\nu_t$ channel. (K4) is the classical odd–even (checkerboard) decoupling of pressure on a collocated grid with central differencing, whose standard remedy is momentum interpolation (Rhie & Chow, 1983); a surrogate-side monitor assembled from plain central differences inherits the pathology without the remedy, and we flag it rather than claim the checkerboard as an exact null mode—the border stencils mean it is not one.

### 6.5 Ranking survives; certification survives only to a floor-limited width

**What the floor costs certification, and why it does not cost triage.** Certification *from the residual norm alone* asks for a map from $\|R_h(\hat{u})\|$ to a bound on $\|\hat{u} - u^\star\|$. The floor does *not* make such a bound invalid—split conformal returns one whatever the score, and ours attains 0.891–0.895 coverage against a 0.90 target. What the floor costs is the bound's **width**. Because $u^\star$ itself is rejected at every threshold below $\|r^\star\|$ (Theorem 2(i)) and the sublevel sets are unbounded in the directions of Proposition 3, a fixed, irreducible part of every score is present at zero error: the median truth-to-prediction residual ratio on the deployed arm is 0.864, so 86% of a typical score is floor. The bound that results is 5.6–6.8× the median drag error it certifies and 1.6× the median field error (subsection 5.9). An interval 1.6× the median error is usable; one 6× the median drag error is not a design tool. And refinement cannot recover it, because the floor grows (subsection 5.6). Ranking asks something much weaker and *between* cases: that a case whose prediction is worse tend to carry a larger monitored residual. Nothing above forbids that, and it is measured—$\rho = 0.61$ and AUROC 0.871 on field error, 0.952 on worst-decile drag error (subsection 5.10). We therefore make the ranking claim empirically and do *not* derive it from (8). In particular we do not argue it from a two-sided bound $\sigma_{\min}\|e\| \leq \|Le\| \leq \sigma_{\max}\|e\|$: $\sigma_{\min}$ is the smallest nonzero singular value of a $\sim 4.7 \times 10^4$ by $6.6 \times 10^4$ advection–diffusion Jacobian with a continuum of near-null directions, it is not separable from zero numerically, and a bound resting on it would be vacuous. Equation (8) offers only the heuristic that far from the truth $\|Le\|$ dominates $\|r^\star\|$; that is a regime intuition, not a guarantee, and the guarantee we do ship for the error bound is the distribution-free conformal one, not a residual bound.

**An apparent contradiction, and its resolution.** On the weak dropout-FNO backbone the prediction sits *below* the floor in 80% of cases ($\|R_h(\hat{u})\| = 0.114$ against $\|r^\star\| = 0.192$), which places it in the regime the decomposition calls floor-dominated—yet $\rho = 0.61$ is measured there. We scope that figure deliberately: it does *not* generalise. On the deployed Transolver the sign reverses, the prediction sits *above* the floor at 0.212–0.215, and the inversion rate is only 2.5–7.0% across three seeds (`results/certificates/transolver_inversion.json`). The contradiction is therefore only apparent on one backbone, but it is worth resolving because the resolution is the same in both. The two are statements about different things. The decoupling is *within* a case: holding the case fixed, moving the field to lower residual need not move it toward the truth, which is exactly what subsection 5.7 measures. The detector claim is *between* cases, where both $r^\star$ and $Le$ vary. That raises the obvious worry—that the ranking is a difficulty proxy rather than an error detector—and we test it: conditioning on each case's own floor leaves a partial rank correlation of 0.561 against a raw 0.610 (subsection 5.10). The detector survives the control, and should be read as between-case triage, never as evidence that residual descent within a case moves toward the truth.

**Empirical confirmation.** On 200/200 real AirfRANS test cases the monitored residual of the ground-truth field is substantially nonzero: $\|r^\star\|$ has mean 0.192 (median 0.133) when averaged over fluid cells, the convention of the cited artifact, and 0.191 in the whole-grid norm named above—the two differ only by the geometric factor $(|\Omega_f|/|\Omega|)^{1/2}$, here 0.995. Meanwhile the uniform-freestream field gives $\|R_h(u_\infty)\| = 0$ in every case—the objective strictly prefers a physically wrong field to the truth (`results/certificates/residual_floor_realdata.json`). Because every field on a given case is scored against the same mask, that factor cancels in any comparison, so the counts below are identical under either convention. A trained backbone's prediction (the dropout-FNO of `checkpoints/certificates_deq.pt`, distinct from the Transolver headline) sits *below* the truth floor in 160/200 cases. That is the flattering number, and it does not generalise: on the *deployed* Transolver the same inversion occurs on only 2.5–7.0% of cases across seeds (8/5/5 of 200 for the backbone alone and 12/9/14 for Transolver+DEQ, seeds $0, 1, 2$; `results/certificates/transolver_inversion.json`, `results/review/functional_audit_gate_analysis.json`), so the truth is the best-scoring field on more than 93% of them, and the median gap reverses sign (+0.013 to +0.015 *above* the floor, against −0.024 below it for the dropout-FNO). Both backbones are scored with an identical ruler: the harness reproduces the published dropout-FNO row to all digits. The inversion is a property of that one backbone and we rest no claim on it. The obvious explanation does not survive measurement: the inversion is *not* over-smoothing. The accurate backbone is not smoother than the truth—its velocity-gradient energy is 1.00× the truth's in a near-body band—so smoothing cannot be what separates the two. (The dropout-FNO measures *rougher* still, 1.75× over the fluid, though we lean on that number only lightly: it is a grid-native field whereas the Transolver field is rasterised from the native cloud, so the cross-backbone gradient comparison carries a rasterisation confound that the inversion counts themselves do not.) What does separate them is that the dropout-FNO's monitored residual is nearly case-*independent* (spread 0.034 versus the truth's 0.162 and the Transolver's 0.161), so it neither tracks case difficulty nor clears the floor. That is what (8) predicts: a prediction falls below the floor only when its error is systematically anti-aligned with $r^\star$, whereas unbiased error of any magnitude *raises* the monitored residual—which is why the accurate backbone sits above it. What does not depend on the backbone is the floor itself, which no model enters. The floor is not an artifact of the deployed grid: re-rasterised from the source point clouds at $128^2$–$512^2$ and scored on a region fixed in physical units it *increases* by 2.6×, fitting $\|r^\star\| \sim h^p$ with $p = -0.64 \pm 0.29$ and decaying in 0/16 cases, while a manufactured analytic solution converges on the same monitor at order $2.02 \pm 0.05$ per case (2.06 fitted on the rung means) and, pushed through the identical rasteriser, still *falls* with refinement (fine/coarse 0.65, rising in only 3/16 cases) where the real truth's rises; and a $C^1$ re-rasterisation *raises* rather than lowers the floor (subsection 5.6). The manufactured field is a potential flow, so the monitor is structurally blind to the inconsistency on it; these controls bound the pipeline's own error, and are not evidence that the monitored operator is consistent. Those band-restricted magnitudes are a different scoring region from the 0.192 above and are not comparable to it; the refinement claim is a statement about the trend within one scoring policy.

**Robustness to the boundary term.** One might object that the monitored $R_h$ omits the no-slip term, so the negative is merely an artifact of dropping the boundary constraint. We tested this with the *framework's own* boundary-inclusive residual $\rho_{\text{res}} = \sqrt{r_c^2 + r_x^2 + r_y^2 + r_{bc}^2}$ (the trust-map definition, $r_{bc}$ the proximity-weighted no-slip penalty) along the correction sweep (dropout-FNO+DEQ, $n$=10, same regime as Table 6; `results/control/bc_inclusive_sweep.json`). The negative *survives*: as the corrector cuts field error (`mse_u` $3.17 \to 2.32$), the boundary-*inclusive* residual rises ($0.145 \to 0.811$) in lockstep with the boundary-excluded one ($0.125 \to 0.807$)—along the correction path the prediction and the truth both approximately satisfy no-slip, so $r_{bc}$ stays small and roughly constant and the interior floor dominates. The objection generalises—*some* no-slip weight might rescue the residual—so we sweep it. Writing the weighted monitor $\rho_\lambda = \sqrt{\|R_h\|^2 + \lambda\,\overline{r_{bc}^2}}$, both logged components are recoverable at every sweep point, so $\lambda$ can be swept post hoc at no computational cost (`scripts/bc_weight_sweep.py`, `results/control/bc_weight_sweep.json`).

**(A)** *At the deployed resolution* the uniform field remains a spurious minimum: it has both a smaller interior residual and a smaller no-slip term than the truth ($\overline{r_{bc}^2} = 0.0053$ versus 0.0073; the ordering holds in 14/16 cases of the resolution study), hence $\rho_\lambda(u_\infty) < \rho_\lambda(u^\star)$ for all $\lambda \geq 0$ *at that resolution*. This leg is resolution-contingent and we claim no more. The framework's proximity weight decays over $3\min(dx, dy)$ and so scores a physically *thinner* band as the grid refines, and the margin collapses monotonically—+9.3% at $128^2$, +7.4% at $181^2$, +6.9% at $256^2$, +3.9% at $362^2$, and −2.8% at $512^2$ in a 16-case run, where it has crossed. A 24-case run over $128^2/256^2/512^2$ agrees on the collapse but still has the truth above by 1.5% at the finest rung (`results/floor_resolution_ladder.json`); a ±2% disagreement between two subsets at a margin that has already collapsed to ∼2% is subset noise, and we take the cautious reading. Where it crosses, the crossover weight $\lambda^\star = \|R_h(u^\star)\|^2 / (\overline{r_{bc}^2}(u_\infty) - \overline{r_{bc}^2}(u^\star))$ has median 599 at $512^2$, the rung where it crosses. Since the deployed trust map uses $\lambda = 1$, a monitor that rescued the ordering would be one in which the no-slip term outweighs the entire PDE residual by two orders of magnitude—a real technical break, but not a practical rescue, because such an object is no longer a PDE monitor. Two further facts close off readings of (A) that would otherwise be live: the far-field ring contributes under 1% of $\overline{r_{bc}^2}(u^\star)$, so the ordering is genuinely about the near-wall band and not an artifact of the crop's outer boundary; and the ordering was never universal, holding at all five rungs in only 11/16 cases.

**(B)** Along the correction path, where field error falls ($3.17 \to 2.32$), *both* increments are positive—the interior term by +0.635 and the no-slip term by +0.0019—so $\rho_\lambda$ rises for every $\lambda \geq 0$; there is no crossover weight. Leg (B) does not depend on the truth-versus-uniform ordering of (A), and it is leg (B) that carries the argument for Table 6: up-weighting the boundary condition cannot convert this residual into a usable correction objective, because the no-slip violation grows along the very path that reduces the error.

**Assumptions, stated plainly.** Theorem 2 is about the *monitored discrete* operator $R_h$, not the continuous PDE. Theorem 1 is the bridge between them, and it is where the smoothness hypotheses live: it holds on a compact subset of the open fluid domain on which the solution is $C^4$ and $\nu_t$ is $C^1$, which excludes a neighbourhood of the wall—the same methodological restriction the refinement ladder makes when it scores a region fixed in physical units. (H2) therefore holds here for a reason that is *not* grid resolution. It follows from operator provenance: the floor grows under refinement, the individually grid-converged convective and pressure-gradient terms ($0.183 \to 0.199$ and $0.180 \to 0.181$) fail to cancel under a discretisation different from the one that produced the labels, and restoring the omitted stress divergence moves the floor by under 0.1%. One further bound: the measured $r^\star$ is evaluated on a *rasterised* $u^\star$—a linear interpolant of an unstructured finite-volume solution—which is not $C^4$ and carries its own representation error, so the measured floor mixes the theorem's term with that error. That is why the theorem is stated as a limit on the exact solution and the magnitude is reported as a measurement, and why we never present one as evidence for the other.

**Relation to prior work.** Krishnapriyan et al. (2021) tie ill-conditioning to the *optimisation difficulty* of PINN losses, and Wang et al. (2022) analyse spectral bias and loss-term imbalance in PINN *training dynamics*; both concern trainability, not a displacement of the residual minimiser from the truth. The forward-versus-backward error gap is classical (Trefethen & Bau, 1997), and a-posteriori PINN bounds upper-bound error *by* residual—the benign regime, and one calibrated by the identity $R(u^\star) = 0$, which is exactly the premise our monitor lacks.

The sharpest way to state where the novelty sits is through goal-oriented estimation. Dual-weighted-residual methods (Becker & Rannacher, 2001) exist precisely because residual magnitude is not an error estimate, and they work beautifully for reduced-order models—because there the residual is $r = A_{\mathrm{FOM}}(u_{\mathrm{ROM}}) - b$, which vanishes at the truth *by construction*, since the full-order operator is available and is the one whose root the reference is. A dataset-trained surrogate has no $A_{\mathrm{FOM}}$. The obstruction we report is therefore about **operator availability**, not about RANS, not about two dimensions and not about the grid: it appears whenever the operator one can afford to evaluate at deployment is not the operator that generated the labels. We do not claim that a dual-weighted estimate provably fails here—that would require constructing the adjoint $z$ and bounding $\langle z, r^\star \rangle$ from below, which we have not done—only that the identity it relies on acquires an uncontrolled $\langle z, r^\star \rangle$ term, and that forming the adjoint needs a linearised solve on the deployed grid, which is the cost the surrogate exists to avoid.

The nearest classical relatives are worth naming precisely, because each has the structure we need and none has the case. Defect correction (Stetter, 1978) and multigrid $\tau$-correction (Brandt & Livne, 2011) both drive one operator using the residual of *another*, and both handle the resulting mismatch by carrying an explicit correction term—exactly the object a deployment-time monitor cannot form, since it would require the fine operator we are trying to avoid running. Zhang et al. (2026) reach the same practical conclusion from the data side for shock physics, and their remedy—defining the loss relative to the residual present in the ground-truth data—likewise requires $R_h(u^\star)$, which a deployment-time monitor by definition does not have. In the finite-element a-posteriori literature, an error contribution the estimator cannot see and that does not vanish under refinement is *data oscillation* (Morin et al., 2000); our floor is its analogue for an inconsistent monitored operator, with the difference that data oscillation is a property of the right-hand side's representation while ours is a property of the operator itself. Least-squares finite element methods (Bochev & Gunzburger, 2009) obtain a residual functional that *is* norm-equivalent to the error—the property our monitor lacks—by constructing the discretisation and the functional together, which is precisely the option a surrogate-side audit forgoes when it adopts a cheap operator after the fact. And where the monitored operator *is* the solver's own, residual-driven correction works on steady CFD (Lei et al., 2026), which locates our negative rather than contradicting it.

That a residual blind to certain modes makes the truth a non-minimiser is, in its bare form, elementary—the pressure gauge above. Our contribution is threefold. **(a)** A *proof* that this floor has a nonzero continuum limit (Theorem 1), so it is a property of operator provenance rather than of the mesh and cannot be refined away even in principle—the $\nu_{\mathrm{eff}}\nabla^2 u$ simplification of the RANS momentum equation is the standard choice in physics-informed CFD surrogates, so this is a floor an entire practice has not accounted for. **(b)** An exact characterisation of the undetectable subspace (Proposition 3)—a counting bound of at least a quarter of the fluid state, exact nonlinear invisibility of the solid, a diagonal $\nu_t$ block with closed-form entries, and the Nyquist pressure mode. **(c)** The demonstration on a learned neural-operator CFD corrector on AirfRANS that the resulting monitor can rank predictions, cannot be descended, and certifies them only at a width the floor rather than the model sets.

## 7 Limitations

- **Backbone-dependent correction quality.** The supervised DEQ corrector reduces field MSE by 8–25% on the SOTA Transolver backbone (subsection 5.2) but is flat-to-unhelpful on the weak grid backbone (subsection 5.3). We report both regimes; we do not claim the corrector helps on an arbitrary backbone, only that it helps on a strong one. The trust signal, by contrast, is positive across all three backbones tested ($\rho = 0.611$ SOTA Transolver, $\approx 0.40$–$0.83$ weak grid backbone, 0.851 bare MeshGraphNet GNN)—but the third of those is not a like-for-like point: that model was trained on 16k-point subsamples and evaluated at $\approx$ 180k, and our own density control returns DENSITY-DRIVEN, so the matched-density range is $\rho \approx 0.40$–$0.61$ (subsection 5.2). We make no competitiveness claim for the grid backbone (it trails SOTA $\sim$ 4–60$\times$); the Transolver backbone we run *on* is itself competitive.

- **The resolution ladder has its own ceiling.** The floor's growth is measured over $128^2$–$512^2$, and at the finest level the raster spacing is only 1.27$\times$ the source cloud's far-field nearest-neighbour spacing, so a still finer level would begin differentiating the interpolant rather than sampling the solution. The $128^2 \to 256^2$ leg is clear of that limit and already shows the rise, but we cannot extend the ladder further on this data and do not claim an asymptotic rate—only that the floor does not decay over the range a $128^2$-trained surrogate is deployed in.

- **The $n$=24 descent arms are Adam, not gradient flow.** The 24-case arms of subsection 5.7 minimise the residual with a first-order optimiser and per-channel scaling, so the field error rises overall but not monotonically and those claims are about endpoints rather than the shape of the path. The $n$=200 arm closes this: it uses Armijo-line-searched gradient descent, under which $J$ is monotone non-increasing by construction, and reaches the same verdict on the deployed backbone under three separate step rules.

- **The headline descent arm freezes $\nu_t$.** Descending the eddy viscosity alongside $(u, v, p)$ quarters the velocity damage, but it does so by dumping the residual into the closure: on the deployed backbone (seed 0) `mse_nut` rises seven orders of magnitude ($6.3 \times 10^{-9} \to 0.115$) while the field error still worsens on 82.5% of cases. We therefore freeze $\nu_t$ in the headline arm and report the four-channel arm as a control.

- **Grid resolution and near-wall blindness.** A uniform $128^2$ Cartesian grid cannot resolve a Re $\approx 10^6$ boundary layer (sub-cell), so wall quantities are approximate and the absolute MSE values are not comparable to body-fitted solvers. Relatedly, the monitored residual zeroes the solid-adjacent (wall-ring) cells—where the central stencil reaches into the $u = v = 0$ discontinuity and registers spurious values—so the trust signal operates on the bulk and is structurally blind to the immediate near-wall band, precisely where near-wall error concentrates on a high-Re airfoil.

- **Two 2-D datasets; not yet 3-D.** The trust signal is validated on two 2-D datasets—turbulent-RANS AirfRANS airfoils and laminar DeepCFD bluff bodies (subsection 5.4, $\rho = 0.77 \pm 0.12$)—so the phenomenon is not AirfRANS-specific, but all results remain 2-D. Extending the residual verifier to 3-D (AhmedML, Ashton et al., 2024; DrivAerNet++, Elrefaie et al., 2024) requires 3-D residual operators and multi-TB, often unsteady datasets beyond a single-GPU budget; we leave it as genuine future work rather than a near-term claim.

- **Conformal coverage out-of-distribution is empirical, not guaranteed.** The distribution-free guarantee assumes exchangeability, which fails under shift; OOD coverage holds in our experiments only because the uncertainty inflates appropriately (subsection 5.9). Scores are also pooled over spatially-correlated cells, so the effective sample size is below the raw cell count, and the calibration set is small ($n = 100$); a larger-calibration-set robustness check is future work.

- **The residual fails as an objective; the gate does not.** Used as a correction *objective* the residual fails: descending it from the exact ground truth drives the field error up on 200/200 cases and on 94–98% of cases started from the deployed backbone (subsection 5.7), and cutting field error along the corrector's own iteration path buys no residual reduction (subsection 5.5). The DEQ contraction guarantee is in the correction variable $\delta$, not in the output's PDE residual. The *acceptance gate* built on the residual is a different mechanism: measured, it admits a step on 99.8% of deployed cases and that step lowers true error on 89.3% of them (subsection 5.5)—though that measurement is a single gated step on the deployed corrector, not the multi-iteration feed-forward loop, which remains untested at this granularity. Nor is the 8–25% field-MSE gain of subsection 5.2 attributable to the residual as a corrector *input*: zeroing that input at train and inference (5 seeds) matches or slightly exceeds the deployed corrector ($\approx$ 4% on a $\approx$ 0.12 base; WITH beats NULL on 0/5 seeds; `results/control/w1_capture.json`), with one caveat stated in section 3—those input channels are built from the wider differentiable stencil, whose larger kernel is a live alternative explanation we have not excluded. The gain is the learned correction's; the residual's demonstrated value is as a case-level trust signal (*which* predictions to distrust), not as a correction input (*how* to fix them).

- **Adaptive uncertainty on the SOTA model requires a deep ensemble.** On the near-deterministic Transolver backbone, MC-dropout $\sigma$ at dropout 0.05 is near-degenerate, so although conformal coverage holds (0.902, target 0.90), the quantile $q$ is enormous ($\approx 10^9$) and ECE $\approx 0.31$: coverage there is delivered by a near-constant band. A deep ensemble of $M$=5 members recovers a genuinely input-adaptive band on this model ($q = 2.35$, ECE 0.074; Table 3), at 5× the training cost. That ensemble certificate comes from a single ensemble training run; the calibration/test split is no longer the caveat (20 re-draws: coverage 0.899±0.009, subsection 5.9), and neither is the choice of $M$: sweeping *every* member subset of size $M = 2..5$ (26 subsets; `scripts/run_ensemble_mstudy.py`, `results/uq_ensemble/mstudy.json`) shows the *decision* layer is insensitive to ensemble size—fused AUROC 0.905–0.912 and conformal coverage 0.905–0.915 (pressure, 0.90 target) at every $M$, so even $M$=2 supports triage—while band *adaptivity* is what $M$ buys: the pressure quantile inflates $2.35 \to 3.74 \to 9.98$ and ECE degrades $0.074 \to 0.113 \to 0.199$ as $M$ drops $5 \to 3 \to 2$ (coverage holds by widening, not by adapting), with $M$=4 already close to $M$=5 ($q = 2.75$, ECE 0.087). Independently retrained ensembles (beyond member subsets of one training run) remain future work.
- **Eddy-viscosity accuracy is backbone-dependent.** Measured against $\nu_t$'s own fluid-masked variance on the AirfRANS test split, the channel is well predicted on the grid backbone ($R^2 = 0.996$) but only moderately on MeshGraphNet ($R^2 = 0.67$–$0.70$; `scripts/check_nut_learned.py`). Because $\nu_t \approx 5.2\,\nu$ on this data and supplies $\approx 84\%$ of $\nu_{\text{eff}}$, a backbone with a weak $\nu_t$ carries that error straight into the monitored residual—so the audit's own operator is only as turbulent as the closure the surrogate predicts. Sharpening $\nu_t$ on the weaker backbones is future work.
- **Classical fallback is a stub interface.** The trust-gated fallback fixes the integration seam and reports what would run; the OpenFOAM/SU2 backends are not implemented and bear on no quantitative claim.
- **The iteration sweep supports decoupling, not divergence.** The sweeps of subsection 5.5 are single-checkpoint and unseeded, and the checkpoints needed to seed them no longer exist, so that $n = 1$ dependency cannot be removed without retraining. A five-seed re-run on the Transolver+DEQ arm (`scripts/iters_sweep_seeded.py`, 24 cases) returns DIRECTION-ONLY against a bar declared in advance: the residual is higher at $k$=15 in 5/5 seeds, but by only +1.5%, at 1.3× the across-seed standard deviation where we required 2×; it is monotone in 0/5 seeds; and paired per case it is higher in 45/120 = 38%, below chance. The invalidating control passes—the fixed-point cap binds in 100% of cases at $k \leq 10$, so the flat tail is a genuine equilibrium rather than a stalled solve. We therefore claim only that cutting field error by 9.4% buys *no* residual reduction, not that the two quantities move in opposite directions. The unqualified claim rests on subsection 5.7 and on the W1 control (0/5 seeds), neither of which has this dependency.

## 8 Conclusion

We set out to have a neural-CFD surrogate audit itself against the governing equations, and found that the operator such an audit can afford is not the operator that produced its training data. The consequences follow from that one fact.

**The ground truth fails the check.** On 200/200 AirfRANS cases the monitored residual of the exact field is substantially nonzero (mean 0.192) while the physically wrong uniform freestream scores an exact zero, so the objective strictly prefers a wrong field to the truth. This is not an artifact of resolution. Refining from $128^2$ to $512^2$ makes the floor *larger*, in 21 of 24 cases, under a rule fixed before the measurement, and a second independent implementation over five rungs finds it decaying in 0 of 16 cases at $p = -0.64 \pm 0.29$; restoring the omitted closure term moves it by under 0.2%; and a higher-order interpolant on the same reference data raises rather than lowers it. What sets the floor is the mismatch between an affordable Cartesian monitor and the body-fitted discrete balance the reference actually solved, and refinement resolves more of that mismatch, not less.

**So the residual cannot be minimised, in a precise sense.** Descending it from the *exact* ground truth—no network anywhere in the loop, so no appeal to an over-smoothed surrogate can explain it away—drives the field error up on **200/200** cases under Armijo-line-searched gradient descent, ending at 1.57× the entire error of the deployed model. On that deployed model the error rises on 94–98% of cases across three seeds ($p < 10^{-30}$), unchanged under three step rules and twelve decades of step size, and monotone in the achieved objective: the harder the residual is minimised, the worse the field.

There is an exception, and it is measurable. Residual descent does *not* always harm the field—on a backbone far from the floor it lowers error on 75% of cases—so this is a regime claim, and the regime is measurable in the ratio of a prediction's residual to the truth's, in which the deployed backbone already sits where descent almost never helps. Even there the objective is unusable: the gain is a thirty-fifth of the supervised corrector's, and there is no stopping rule, since the objective falls monotonically while the error minimum is interior on 78.5% of cases. What never happens is that the residual selects the right iterate.

**The boundary is the operator, not the problem class.** Newton–Krylov and related corrections succeed on steady CFD precisely because the residual they drive is the solver's own, whose floor at the truth is zero by construction. Our negative result and those positive ones are not in tension; they are opposite sides of operator consistency, which is a sharper boundary than the steady-versus-transient or resolved-versus-under-resolved lines it might be confused with.

**What the audit delivers.** A working one. The whole certificate—residual maps, trust map, case-level score and conformal band—costs a median 1.13 ms against a 3822 ms solve, three hundredths of a percent, and rejecting the least-trusted 10% of cases by the fused score recovers ≈91% of the error reduction an oracle could achieve at that budget. The residual is a useful between-case error *detector*, positive on every backbone tested—though on field error a physics-free ensemble score matches it and a paired bootstrap cannot separate the two; the physics wins outright on drag (AUROC 0.952). The detector is not merely a difficulty proxy: conditioning on each case's own floor leaves partial $\rho = 0.561$ against a raw 0.610. The split-conformal layer holds target coverage in and out of distribution, and needs no operator at all; where the residual is the score being conformalised, the certificate stays valid (0.891–0.895 at a 0.90 target) but at a width the floor rather than the model sets—5.6–6.8× the drag error it bounds. The monotone-residual gate is a genuine guarantee—it cannot raise the monitored residual—but its *accuracy* advantage does not survive an ungated fixed half step, so we claim the certificate and not the accuracy. And the learned corrector that improves a competitive backbone is supervised toward truth, with a controlled ablation attributing its gain to the learned correction rather than to the residual it ingests.

The durable contribution is therefore a measurement and the boundary it draws, not a system: an audit built on a proxy operator can tell you *which* predictions to distrust, and cannot tell you how to fix them, because the field it would move you toward is not the one you want. Anyone building a surrogate-side physics check should measure their operator's floor on their own ground truth first; if it is not near zero, the detector may still be worth having, but the objective is not.

## Code and data availability

The complete source, training/evaluation harnesses, and per-claim reproduction map (`docs/REPRODUCE.md`) are released as the open-source package `neuroforge-cfd` (https://github.com/ali-kin4/neuroforge-cfd; archived at Zenodo, DOI 10.5281/zenodo.21277928). Every headline number maps to a committed script and result file, and a top-level manifest (`results/MANIFEST.json`) records seeds, environment, and SHA-256 hashes for hash-drift detection. The two benchmarks are public: AirfRANS (Bonnet et al., 2022) and DeepCFD (Ribeiro et al., 2020). The package is CPU-first and runs end-to-end with zero downloads via a synthetic data generator; the GPU runs (backbone training) are documented with exact commands and expected outputs.

## CRediT author contribution statement

**Ali Jabbary:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Kasra Ghanavati:** Validation, Writing – review & editing.

## Declaration of competing interests

The authors declare no competing financial or personal interests.

## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to assist in drafting and editing the manuscript text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. The use of AI assistance in developing the research software is described in the Implementation section.


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